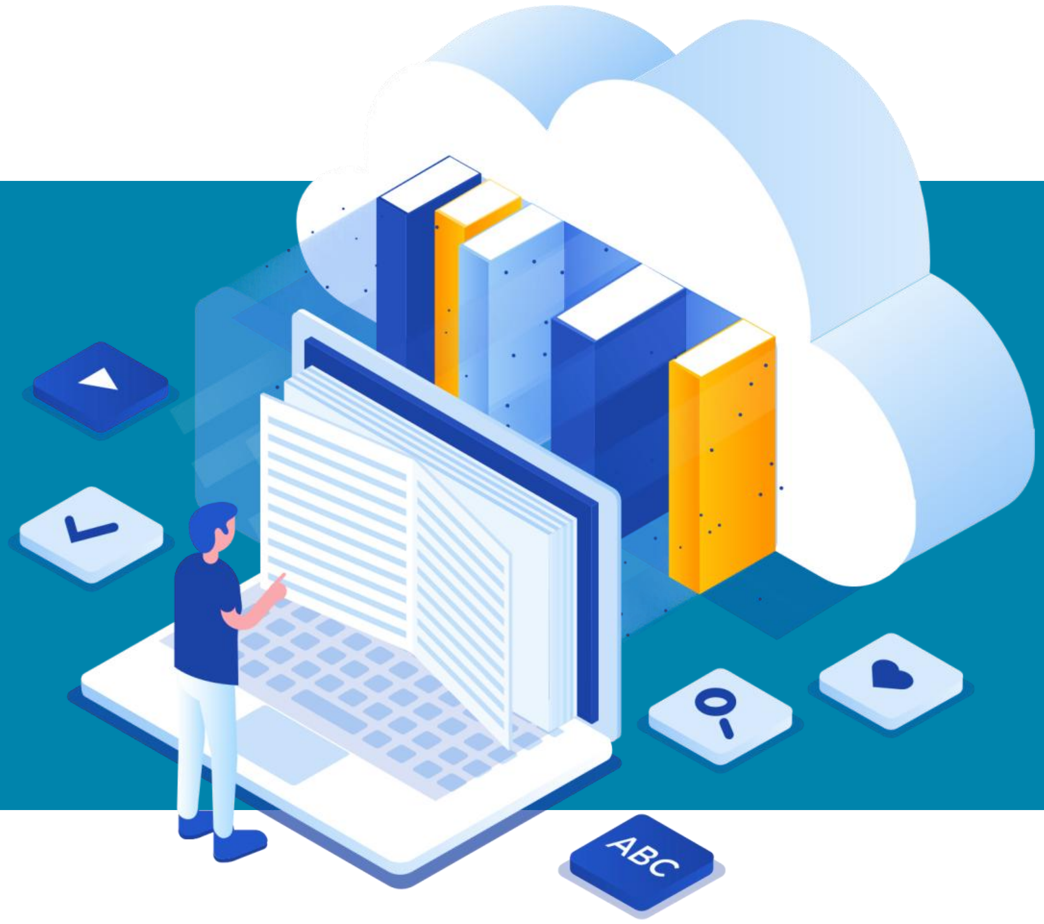


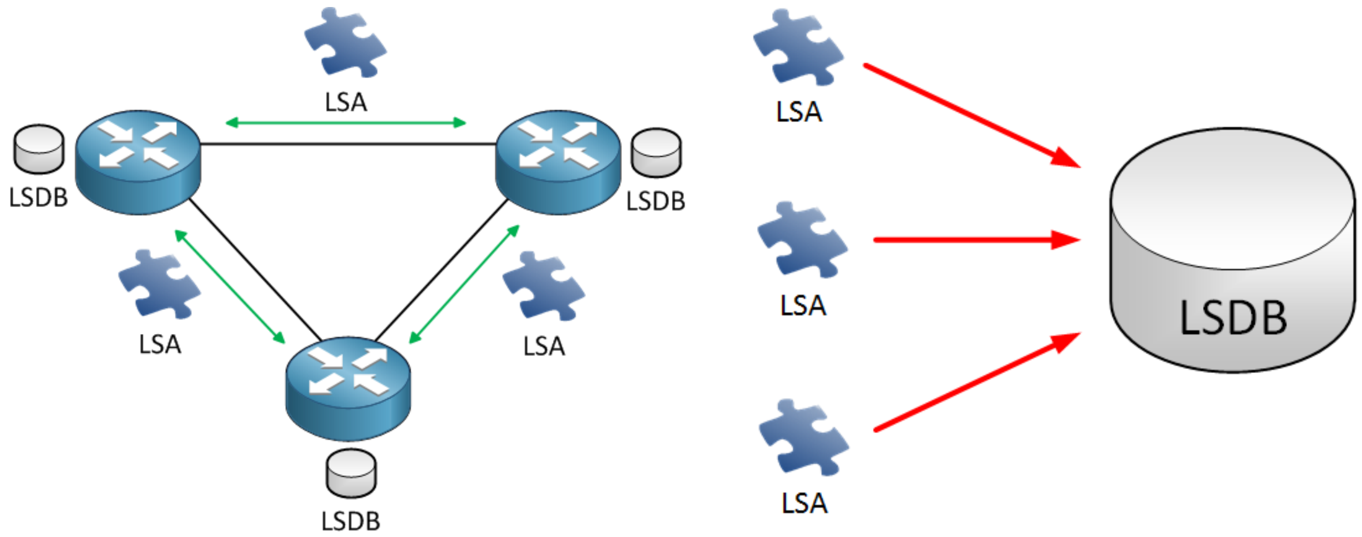
OSPF

What is OSPF ?

- 표준 라우팅 프로토콜 TCP/IP 환경에서 가장 많이 사용되는 라우팅 프로토콜
- Link-State Routing Protocol로 네비게이션 시스템과 유사
  - 완전한 지도를 갖고 목적지에 도달하기 위한 가장 짧은 경로 탐색 (최적 경로)
  - 모든 것을 알고 있다는 것은 Loop이 불가능 하다는 뜻
- 운영하는 장비와 규모가 커지면 CPU 사용률이 높아짐
  - 서울 지도에서 경로 찾기 vs 전국 지도에서 경로 찾기
  - 지도의 크기를 조절하기 위해 Area 단위로 나눠 운영

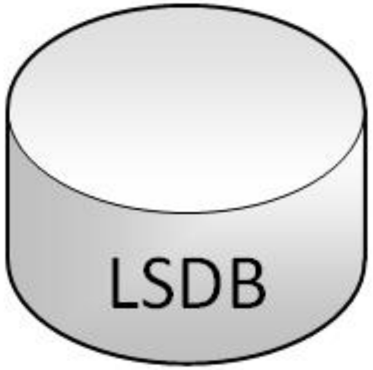
OSPF

- Link : 라우터의 인터페이스
- State : 인터페이스에 연결된 인접 라우터의 연결 상태
- OSPF는 Link-State Advertisements(LSA)를 다른 인접 라우터에게 전송함으로써 작동
- 인접 라우터간 주고 받은 LSA를 이용해 LSDB 구성



What is OSPF ?

- LSDB는 네트워크에 대한 OSPF의 전체 그림
  - 즉. 토폴로지 라 부름
  - LSDB를 국가 전체 지도와 비교할 수 있음
- 모든 라우터가 지도를 갖게 되면 SPF 알고리즘을 이용해 모든 목적지까지 가는 최단 경로 계산
  - 최적 경로는 라우팅 테이블에 등록



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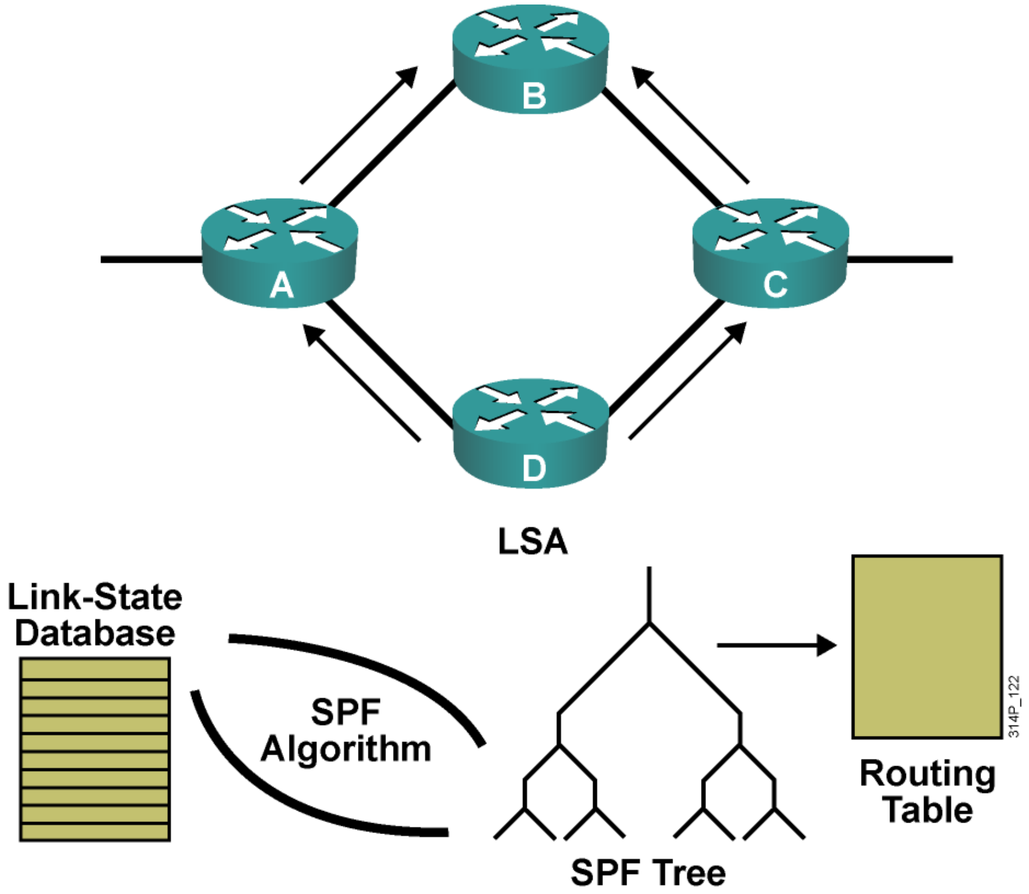
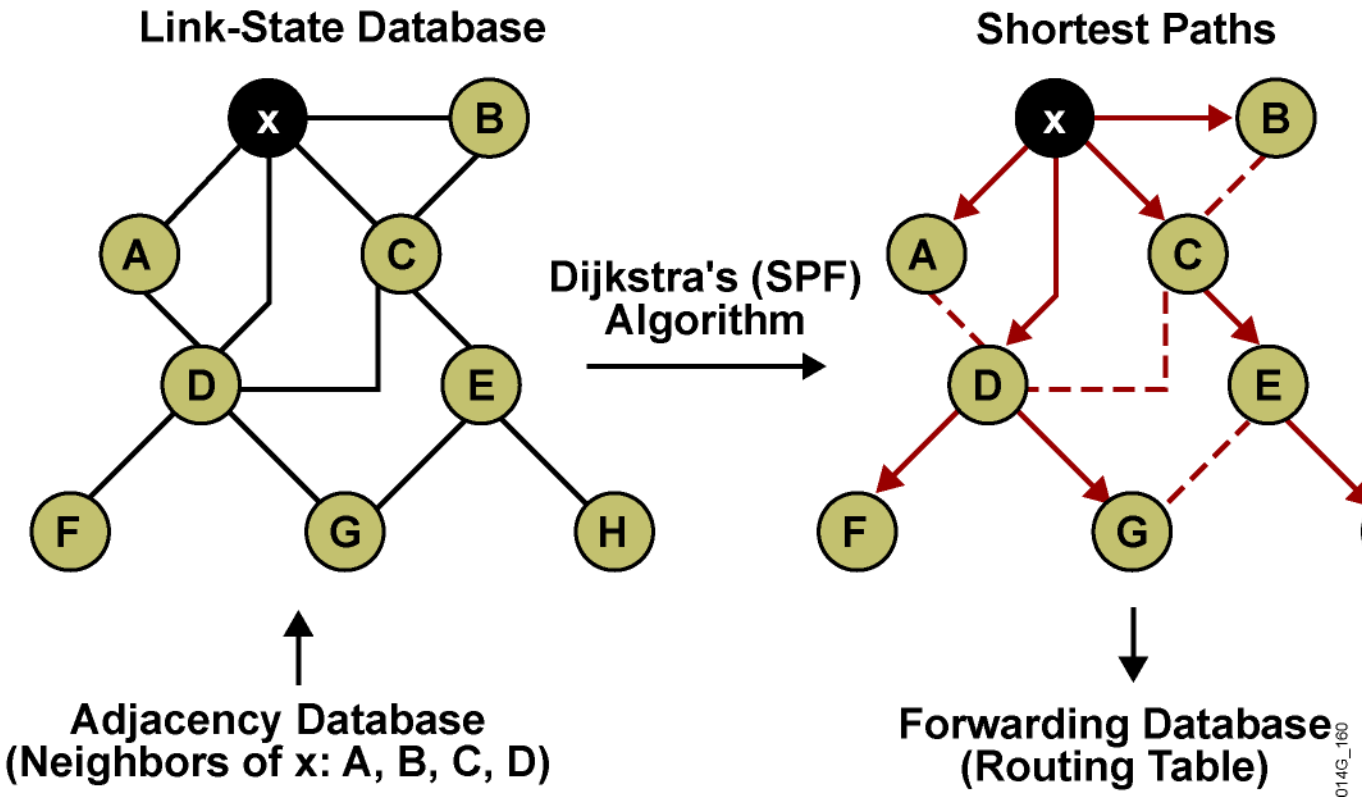
OSPF Summary



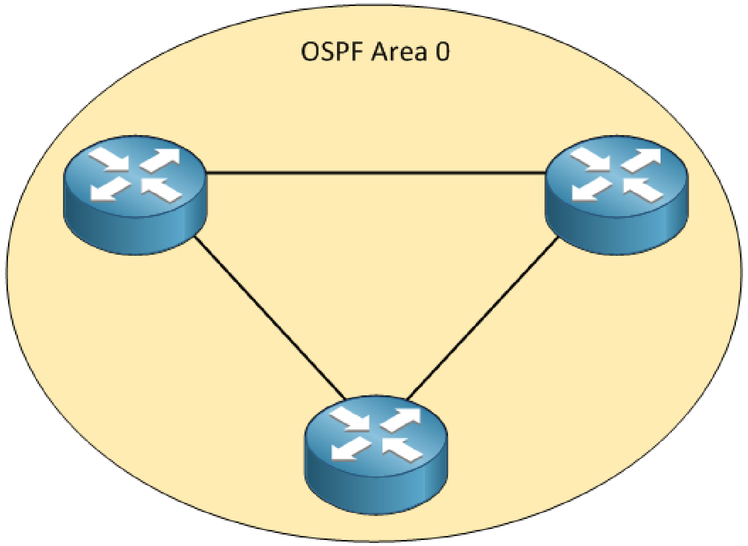
OSPF Summary



Assume all links are Ethernet, with an OSPF cost of 10.

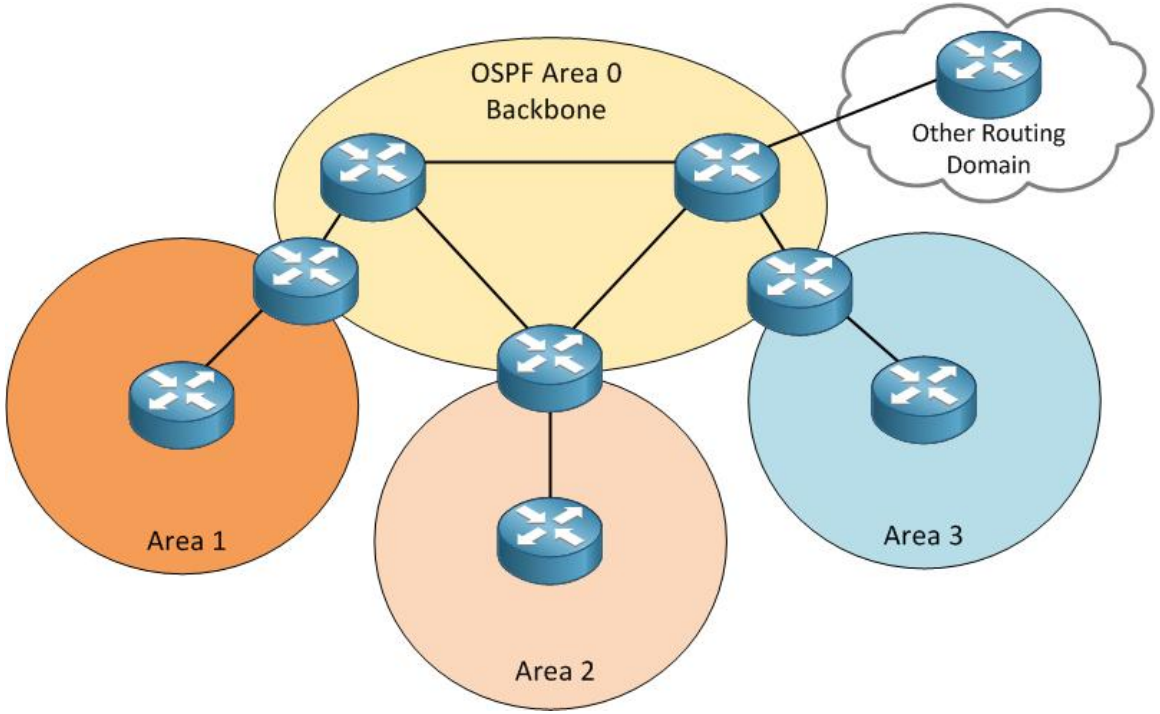
OSPF Area

- OSPF는 Area 개념과 함께 동작
  - 단일 영역 환경에서는 기본적으로 항상 Area 0 동작
  - Area 0 : Backbone Area



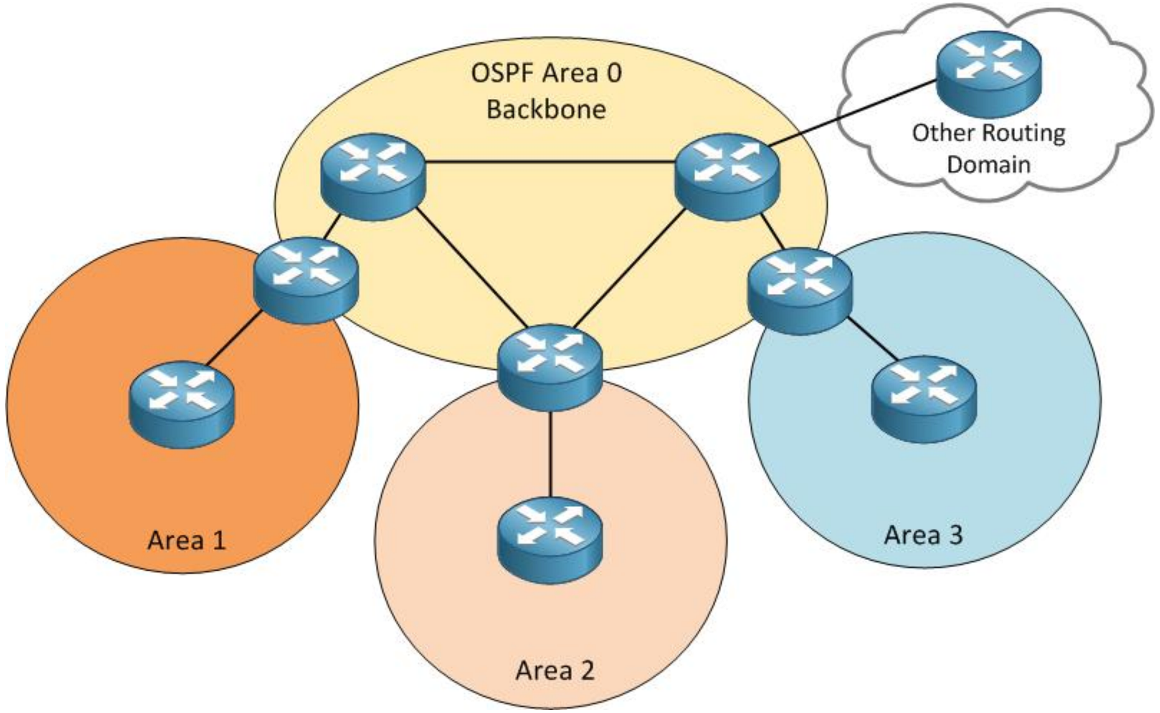
OSPF Area

- OSPF는 여러 Area를 갖고 동작할 수 있지만, 모든 영역은 Backbone Area (Area 0)에 연결되어야 함
  - Area 간 통신은 항상 Backbone Area 통과해야 가능

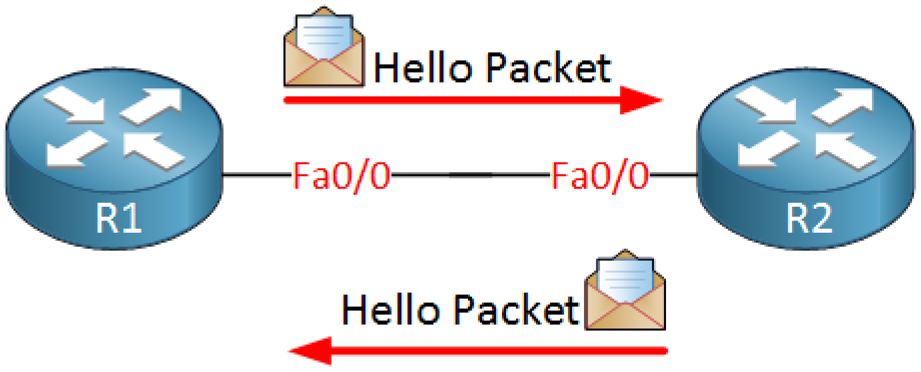


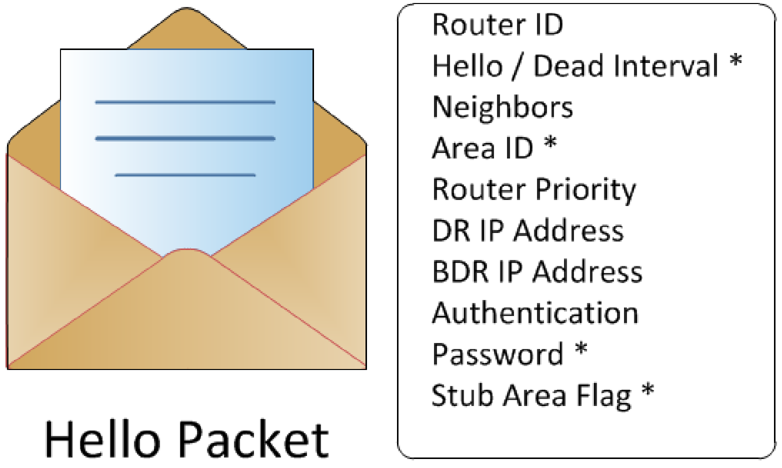
OSPF Area

- Backbone Area에서 동작 하는 라우터 : Backbone Router
- 두개의 Area 사이에서 동작 하는 라우터 : Area Border Router
- OSPF와 함께 다른 Routing Protocol 운영하는 라우터 : Autonomous System Border Router



OSPF Neighbor

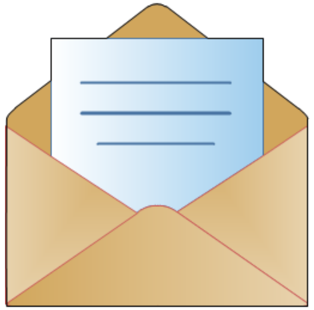




- Link-State 정보를 주고 받기 위해서는 인접 장비와 Neighbor 관계 형성 필요
  - OSPF를 구성하면 라우터는 OSPF 활성화된 인터페이스로 Hello 패킷을 송신
  - 멀티캐스트를 이용 (224.0.0.5)
- Hello Packet 정보 중 몇가지는 반드시 일치해야 Neighbor 관계 형성 가능
  - Hello/Dead time, Area-ID, Authentication Password, Stub Flag

OSPF Neighbor

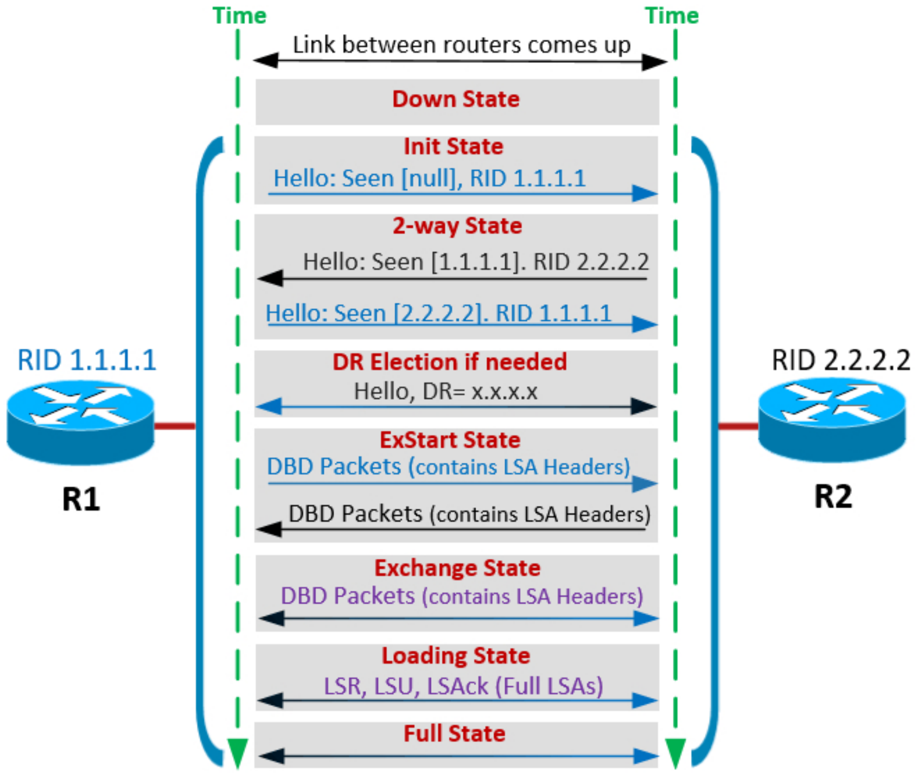
- Router ID : 인접 라우터를 구분하기 위한 고유 ID
- **Hello/Dead time** : Hello 메세지 전송 주기 / 인접 장비의 Dead 선언 시간 (Neighbor 관계 종료)
- **Area ID** : 라우터가 속한 Area
- Router Priority : DR / BRD 선출을 위한 우선 순위 (DR / BDR 선출 환경)
- DR / BDR IP : DR의 IP 및 BDR의 IP (DR / BDR 선출 환경)
- **Authentication Password** : 안전한 Neighbor 관계 성립을 위해 인증 설정 (Option)
- **Stub Area flag** : Stub란 Area와 다른 유형의 Area로, 동일한 Stub Area에 속해야 Neighbor 가능 (Option)



Hello Packet

Router ID
Hello / Dead Interval *
Neighbors
Area ID *
Router Priority
DR IP Address
BDR IP Address
Authentication
Password *
Stub Area Flag *

OSPF Neighbor



- **DOWN**
  : Hello Packet 미수신
  : Deadtime 동안 Hello Packet 미수신
  : 인접 장비와 비정상 연결

- **Init**
  : 단방향 Hello Packet 수신 상태

- **2-way**
  : Neighbor 장비와 쌍방 통신이 이뤄진 상태

- **Exchange**
  : OSPF 설정된 장비들이 DBD 를 교환하는 단계

- **Loading**
  : DBD 수신 후 필요에 따라 LSR,LSU,LSAck 교환하는 단계

- **Full**
  : Neighbor 장비들간 라우팅 정보교환이 끝난 상태

OSPF Router ID

- OSPF는 효율적인 운영 환경을 위해 고유한 Router ID가 필요 (IP 주소 형식)
  1) OSPF 설정 과정에서 운영자가 직접 입력
  2) OSPF는 라우터의 Loopback IP를 Router ID로 사용
     - 라우터의 Loopback 인터페이스가 여러 개 있을 경우 Loopback 중 가장 높은 IP를 Router ID로 사용
  3) 라우터에 Loopback 인터페이스가 없는 경우, 물리적 인터페이스의 가장 높은 IP를 Router ID로 사용
- OSPF Neighbor 협상 과정에서 결정된 Router ID는 변경 불가
  - Router ID 변경을 위해서는 OSPF Neighbor 재협상 필요

```
Router(config)#router ospf 1
Router(config-router)#router-id 172.16.1.1

Router#clear ip ospf process   (* 이미 등록된 Router ID 변경을 위해선 Processes 재시작 필요)
```

OSPF Metric

- OSPF Metric 은 인터페이스의 대역폭을 기반으로 Cost 표기
- Cost 계산법
  - Cost = Reference bandwidth ÷ Interface bandwidth
  - 1 이하는 올림
  - 낮을 수록 빠른 경로
- Default reference bandwidth
  - 10^8 = 100,000,000 bps

Interface	Bandwidth (bps)	Cost
FastEthernet	100,000,000	100,000,000 / 100,000,000 = 1
GigaEthernet	1,000,000,000	100,000,000 / 100,000,000 = 0.1 => 1
TenGigaEtnernet	10,000,000,000	100,000,000 / 10,000,000,000 = 0.01 => 1

* 빠른 interface 사용하도록 유도 하기 위해 reference bandwidth 변경 필요

OSPF Basic Configuration

```
Router(config)#
```

router ospf *process-id*

- Enables one or more OSPF routing processes

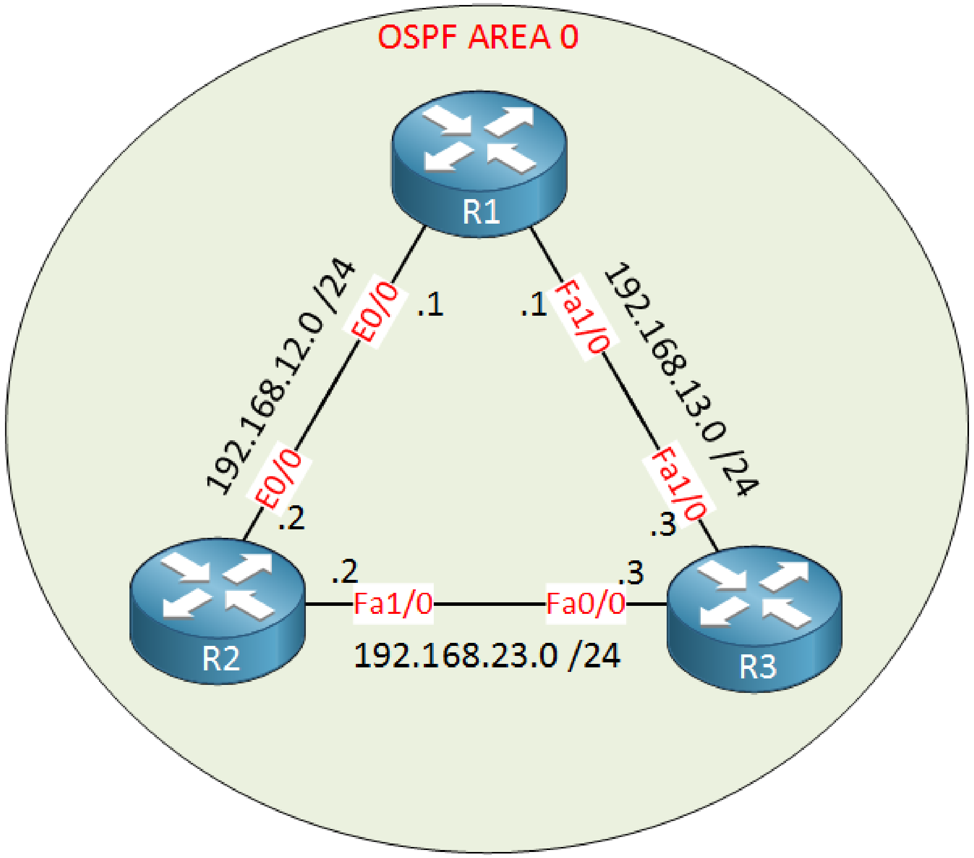
```
Router(config-router)#
```

network *ip-address wildcard-mask* **area** *area-id*

- Defines the interfaces that OSPF will run on

OSPF Basic Configuration

- 모든 라우터는 OSPF Area 0에서 동작




```
R2(config)#router ospf 1
R2(config-router)#network 192.168.23.0 0.0.0.255 area 0
```

```
R3(config)#router ospf 1
R3(config-router)#network 192.168.23.0 0.0.0.255 area 0
```

1) 숫자 '1'은 프로세스 ID로 원하는 숫자 선택 가능
   - ✓ 각 라우터는 다른 프로세스 ID 선택 가능
2) Network 명령어 통해 Network 광고 및 OSPF 활성화
   - ✓ 내가 갖은 네트워크 중 이 범위에 속하는 네트워크를 OSPF 통해 광고
   - ✓ 내가 갖은 인터페이스 중 이 범위에 속하는 인터페이스 통해 Hello 전송
   - ✓ 네트워크 범위 지정은 Wildcard Mask 사용
   - ✓ 이 범위의 네트워크/인터페이스가 속한 Area

OSPF Basic Configuration – Wildcard Mask

- Wildcard Mask는 역 서브넷 마스크
  - 즉. Wildcard Mask의 0 과 1은 서브넷 마스크의 반대

Subnetmask	255	255	255	0
	11111111	11111111	11111111	00000000
Wildcardmask	0	0	0	255
	00000000	00000000	00000000	11111111

OSPF Basic Configuration – Verification#1

- OSPF Neighbor 성립 Log

```
R3# %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.2 on FastEthernet0/0 from LOADING to FULL, Loading Done
```

```
R2# %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.3 on FastEthernet1/0 from LOADING to FULL, Loading Done
```

- OSPF Neighbor 상태 확인

```
R3#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
192.168.23.2 1 FULL/BDR 00:00:36 192.168.23.2 FastEthernet0/0
```

```
R2#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
192.168.23.3 1 FULL/DR 00:00:32 192.168.23.3 FastEthernet1/0
```

OSPF Basic Configuration – Verification#2

- OSPF Process 확인

```
R2#show ip protocols
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.23.2
```

```
R3#show ip protocols
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.23.3
```

OSPF Basic Configuration – Change Route ID

- Router ID는 Neighbor 협상 과정에서 결정 되고 운영 중 변경 불가
  - 재협상 필요

```
R2(config)#interface loopback 0
R2(config-if)#ip address 2.2.2.2 255.255.255.0

R2#show ip protocols
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 192.168.23.2
```

```
R2#clear ip ospf process
Reset ALL OSPF processes? [no]: yes

R2#show ip protocols
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 2.2.2.2
```

OSPF Basic Configuration – Change Route ID

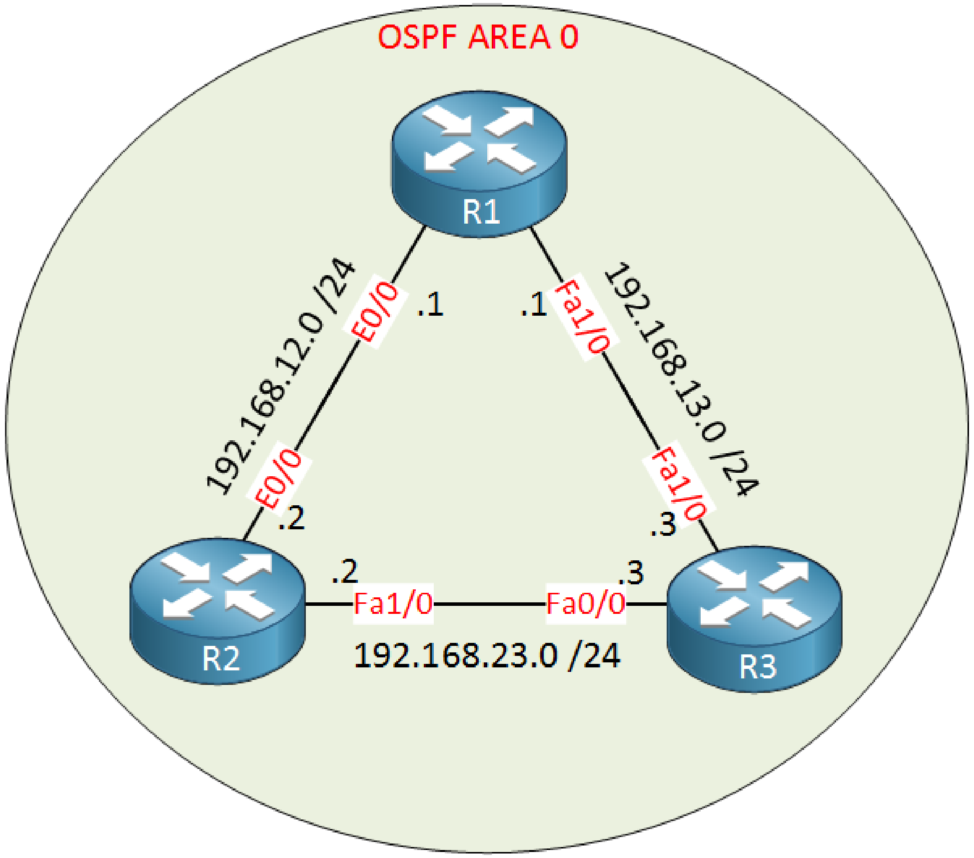
- Router ID는 Neighbor 협상 과정에서 결정 되고 운영 중 변경 불가
  - 재협상 필요

```
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
Reload or use "clear ip ospf process" command, for this to take effect
```

```
R3#clear ip ospf process
Reset ALL OSPF processes? [no]: yes

R3#show ip protocols
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 3.3.3.3
```

OSPF Basic Configuration

- 모든 라우터는 OSPF Area 0에서 동작




```
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 0
```

```
R1(config)#router ospf 1
R1(config-router)#network 192.168.12.0 0.0.0.255 area 0
R1(config-router)#network 192.168.13.0 0.0.0.255 area 0
```

```
R3(config)#router ospf 1
R3(config-router)#network 192.168.13.0 0.0.0.255 area 0
```

OSPF Basic Configuration – Verification#3

- OSPF Neighbor 상태 확인

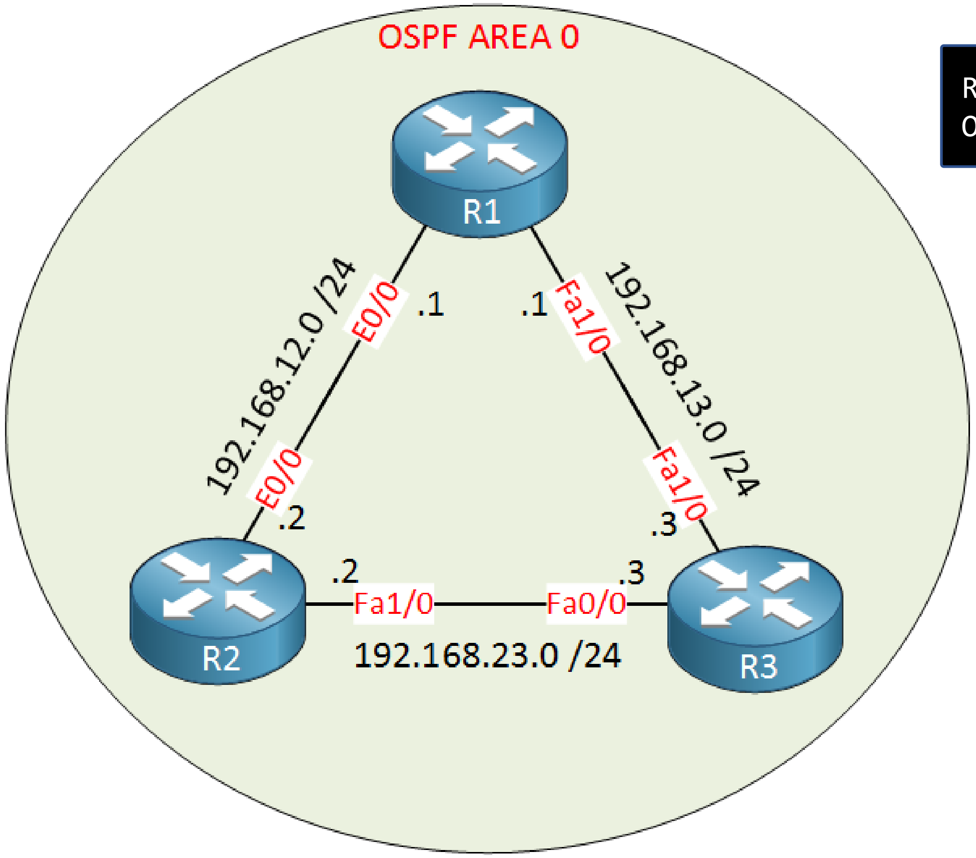
```
R2#show ip ospf neighbor

Neighbor ID     Pri   State       Dead Time   Address         Interface
192.168.13.1      1   FULL/BDR    00:00:31    192.168.12.1    Ethernet0/0
3.3.3.3           1   FULL/DR     00:00:38    192.168.23.3    FastEthernet1/0
```

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State       Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR    00:00:33    192.168.13.3    FastEthernet1/0
2.2.2.2           1   FULL/DR     00:00:30    192.168.12.2    Ethernet0/0
```

```
R3#show ip ospf neighbor

Neighbor ID     Pri   State       Dead Time   Address           Interface
192.168.13.1      1   FULL/DR     00:00:37    192.168.13.1      FastEthernet1/0
2.2.2.2           1   FULL/BDR    00:00:30    192.168.23.2      FastEthernet0/0
```

OSPF Basic Configuration – Verification#4

- Routing Table 확인

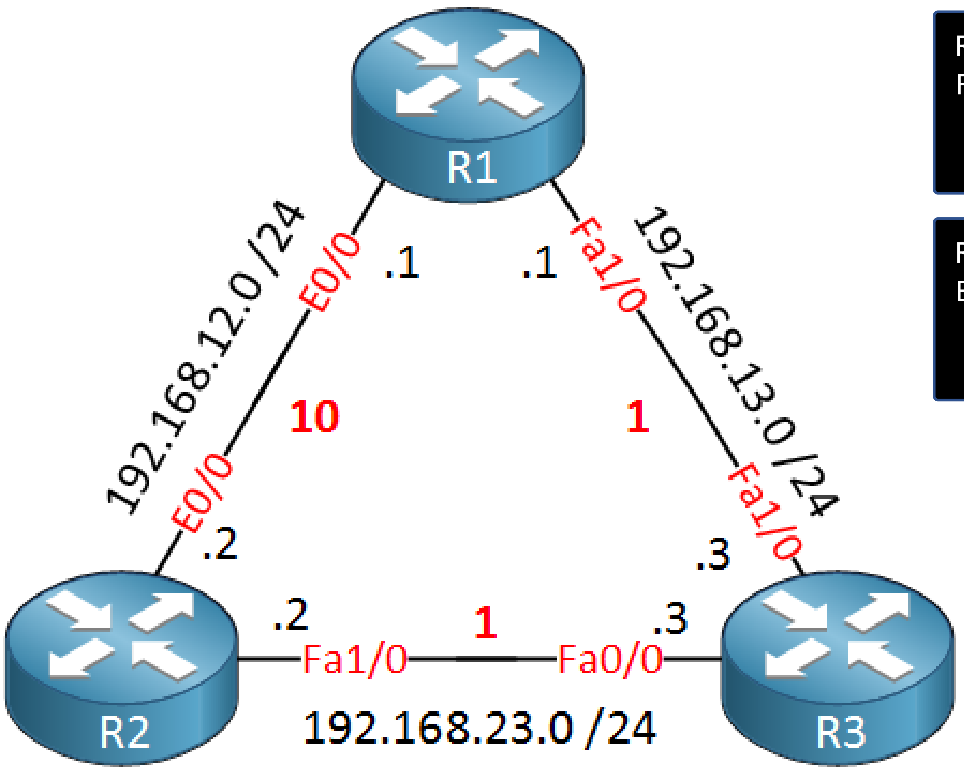



```
R2#show ip route ospf
O    192.168.13.0/24 [110/2] via 192.168.23.3, 00:09:45, FastEthernet1/0
```

1) **'O'** : OSPF를 나타냄, 즉. 이 Network는 OSPF를 통해 학습
2) **'192.168.13.0/24'** : 학습한 Network
3) **'110'** : OSPF의 Administrative distance 값
4) **'2'** : 학습한 Network의 Cost
5) **'via 192.168.23.3'** : 해당 Network에 도달하기 위한 Next-Hop 주소
6) **'FastEthernet1/0'** : Next-Hop에 도달하기 위한 Outgoing 인터페이스
   - 이 네트워크는 Fa1/0에 연결된 OSPF Neighbor 통해 학습

OSPF Basic Configuration – Verification#5

- Routing Table 확인



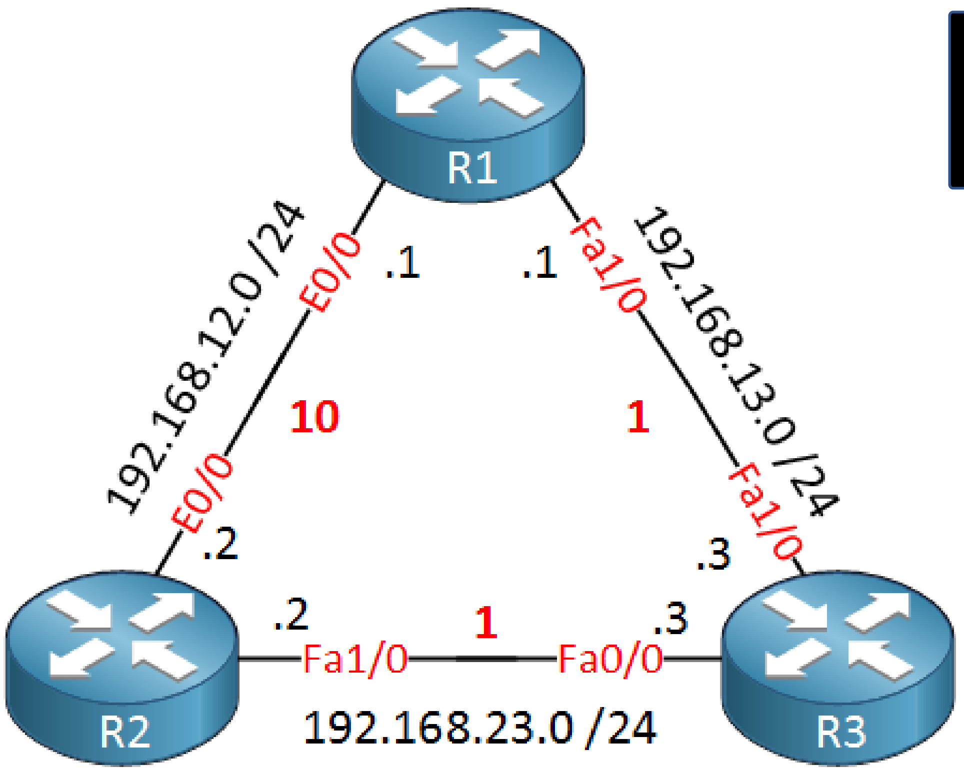

```
R2#show ip ospf interface fa1/0
FastEthernet1/0 is up, line protocol is up
  Internet Address 192.168.23.2/24, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 1
```

```
R2#show ip ospf interface e0/0
Ethernet0/0 is up, line protocol is up
  Internet Address 192.168.12.2/24, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 10
```

- 192.168.13.0/24로 향하는 IP 패킷의 경로
  - ✓ **R3 통해 도달 : 1+1 = Cost 2**
  - ▪ R1 통해 도달 : 10+1 = Cost 11

OSPF Basic Configuration – Verification#5

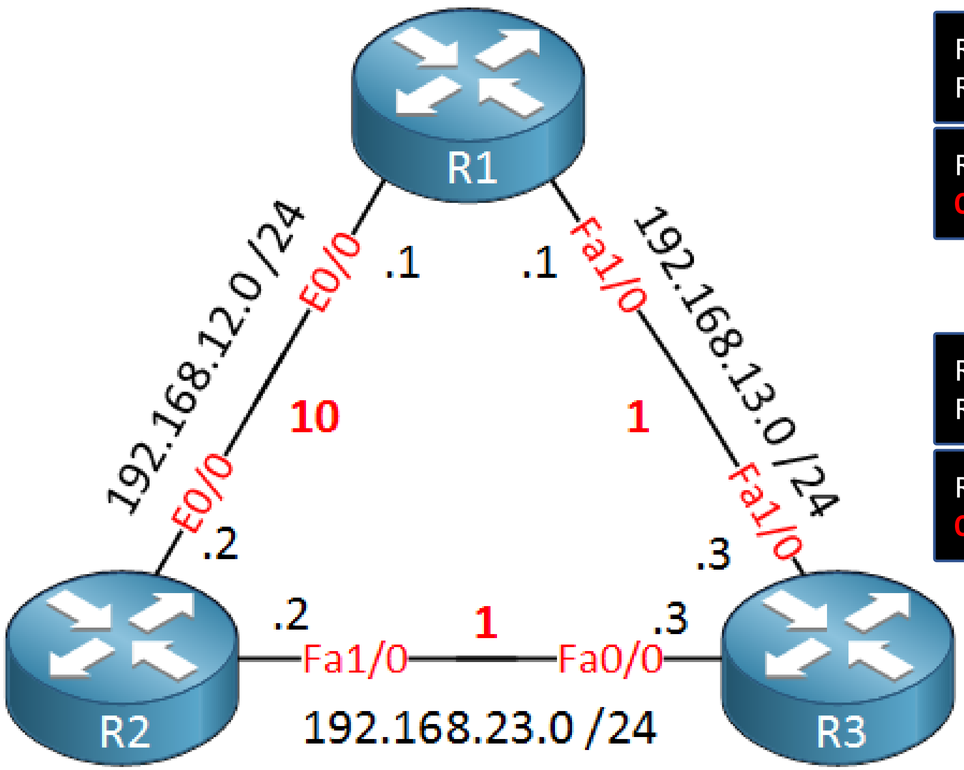
- Routing Table 확인 – LB




```
R3#show ip route ospf
O     192.168.12.0/24 [110/11] via 192.168.23.2, 00:01:14, FastEthernet0/0
                      [110/11] via 192.168.13.1, 00:01:14, FastEthernet1/0
```

OSPF Basic Configuration – Option#1

- Backup 경로 확인




```
R3(config)#interface fastEthernet 0/0
R3(config-if)#shutdown
```

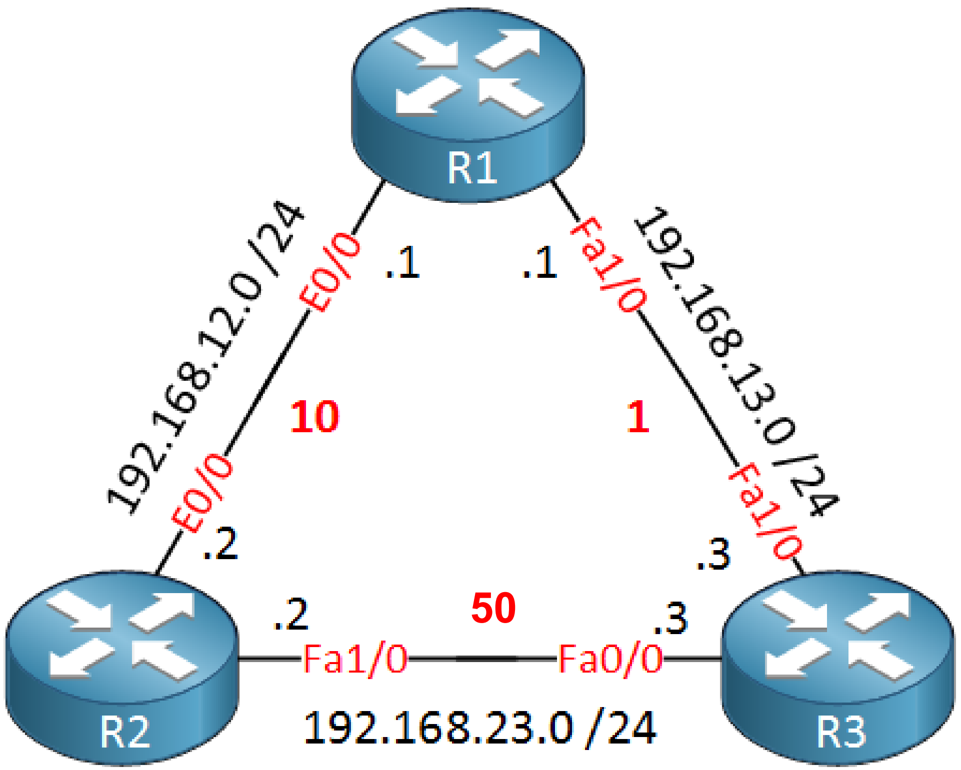
```
R2#show ip route ospf
O     192.168.13.0/24 [110/11] via 192.168.12.1, 00:01:20, Ethernet0/0
```

```
R3(config)#interface fastEthernet 0/0
R3(config-if)#no shutdown
```

```
R2#show ip route ospf
O     192.168.13.0/24 [110/2] via 192.168.23.3, 00:00:01, FastEthernet1/0
```

OSPF Basic Configuration – Option#2

- 경로 Cost 조정 – 경로 선택




```
R2(config)#interface fastEthernet 1/0
R2(config-if)#ip ospf cost 50
```

```
R2#show ip ospf interface fastEthernet 1/0
FastEthernet1/0 is up, line protocol is up
  Internet Address 192.168.23.2/24, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 50
```

```
R2#show ip route ospf
O    192.168.13.0/24 [110/11] via 192.168.12.1, 00:01:20, Ethernet0/0
```

```
R2(config)#interface fastEthernet 1/0
R2(config-if)#no ip ospf cost 50
```

```
R1#show ip route ospf
O    192.168.23.0/24 [110/2] via 192.168.13.3, 00:00:15, FastEthernet1/0
```

OSPF Reference Bandwidth

- Cisco IOS Default reference Bandwidth = 10^8 bps
  - = 100,000,000 bps = 100,000 Kbps = 100 Mbps
- Cost = reference bandwidth / interface bandwidth

Fast Ethernet

```
Router#show ip ospf | include Reference
 Reference bandwidth unit is 100 mbps

Router#show interfaces FastEthernet 0/0 | include BW
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec

Router#show ip ospf interface FastEthernet 0/0 | include Cost
  Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1

100.000 kbps reference bandwidth / 100.000 interface bandwidth = 1
```

Serial

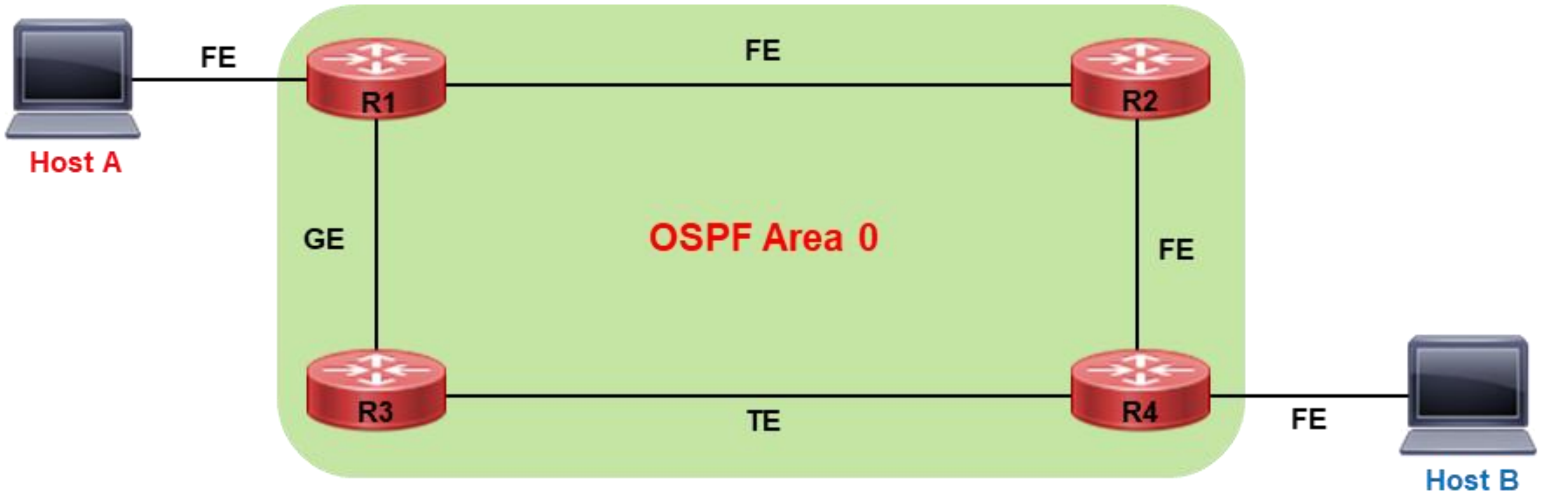
```
R1#show interfaces Serial 0/0 | include BW
  MTU 1500 bytes, BW 1544 Kbit/sec, DLY 20000 usec,

R1#show ip ospf interface Serial 0/0 | include Cost
  Process ID 1, Router ID 192.168.2.1, Network Type POINT_TO_POINT, Cost: 64

100,000 kbps reference bandwidth / 1,544 kbps interface bandwidth = 64.76
```

OSPF Reference Bandwidth

- Default reference bandwidth의 문제점
  - Fast Ethernet = 1Gigabit Ethernet = 10 Gigabit Ethernet = Cost 1
- 1G 이상의 OSPF를 운영하는 환경에서는 reference bandwidth 변경 필요

```
Router(config-router)#auto-cost reference-bandwidth ?
  <1-4294967>  The reference bandwidth in terms of Mbits per second

Router(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers

Router#show ip ospf | include Reference
 Reference bandwidth unit is 1000 mbps

Router#show ip ospf interface FastEthernet 0/0 | include Cost
  Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 10
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
```

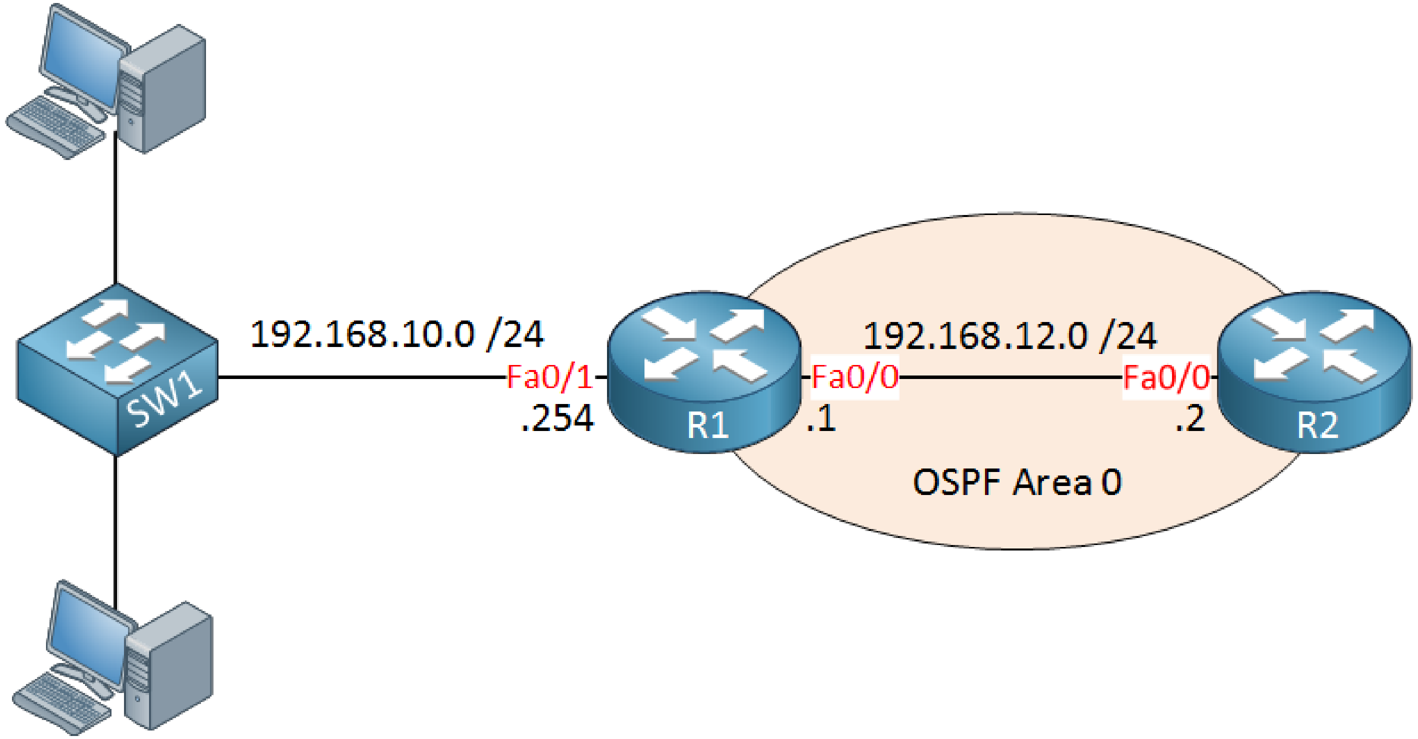
OSPF Reference Bandwidth



Reference Bandwidth (Default)	Reference Bandwidth (auto-cost reference-bandwidth 10000)
R1 → R2 → R4 → Host B (Cost 3)	R1 → R2 → R4 → Host B (Cost 300)
R1 → R3 → R4 → Host B (Cost 3)	R1 → R3 → R4 → Host B (Cost 111)
Load Balancing 적용	R1 → R3 → R4 경로 사용

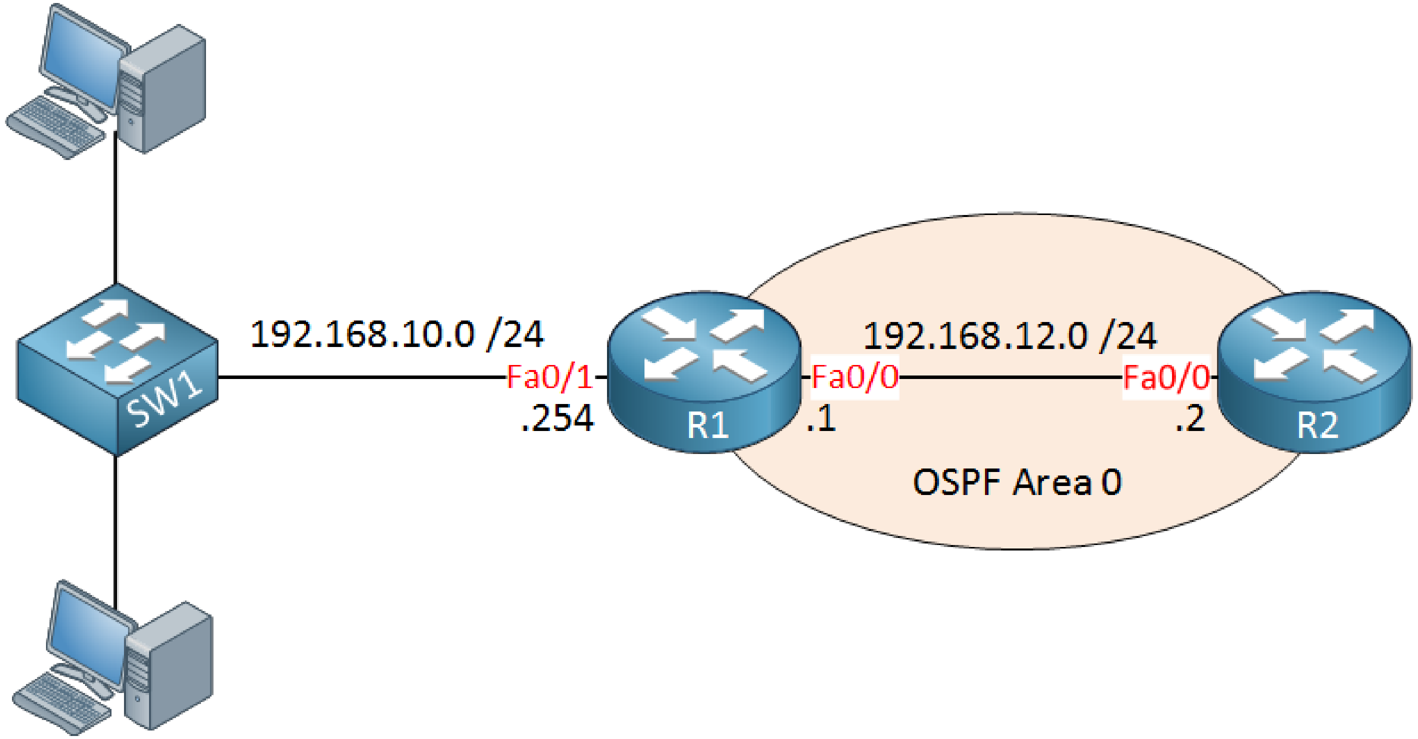
OSPF Passive Interface

- OSPF Network 명령어의 두 가지 동작
  - 네트워크 명령어의 범위에 속하는 Connected Network는 모두 OSPF로 광고
  - 네트워크 명령어의 범위에 속하는 IP가 설정된 인터페이스를 통해 Hello 전송



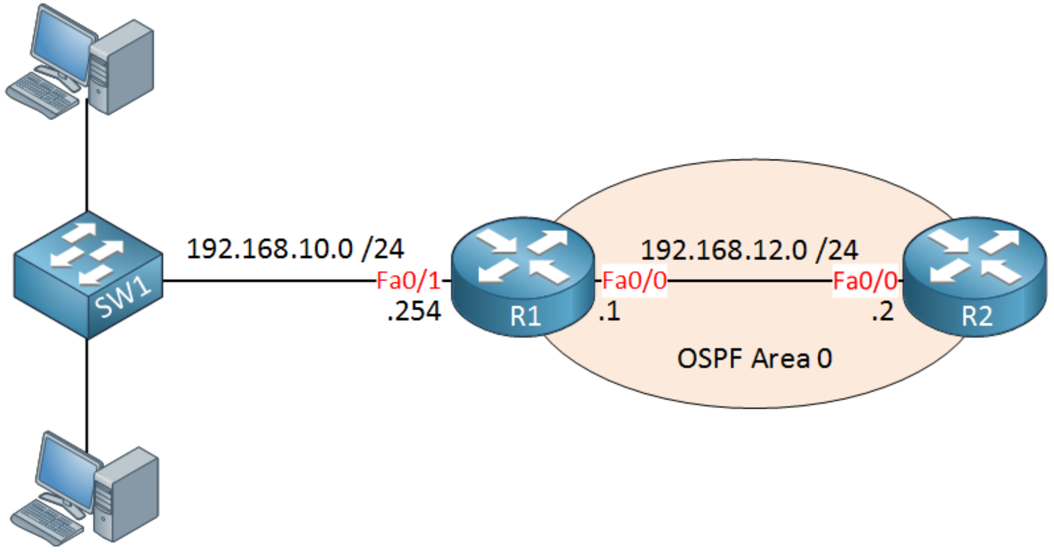
OSPF Passive Interface

- R1이 '192.168.10.0/24' 네트워크를 R2에게 광고 하고자 할 때 Network 명령어 사용
- 이때 R1은 R2에게 '192.168.10.0/24' 경로를 광고 하지만, 동시에 Fa0/1 인터페이스 통해 Hello 전송
  - 누군가 악의적인 의도를 갖고 R1의 fa0/1 인터페이스와 OSPF Neighbor를 맺고 경로를 전달 해 데이터 수집 가능



OSPF Passive Interface

- 특정 인터페이스에 Hello 패킷 전송을 하지 않게 하기 위해 Passive-Interface 사용



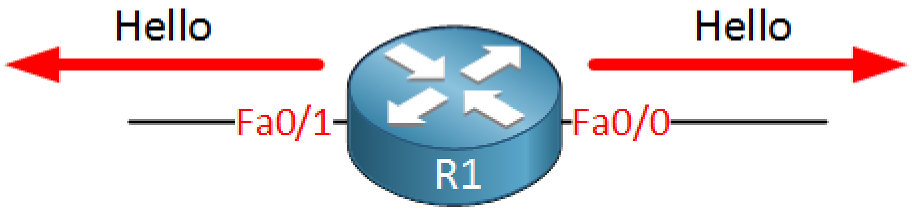

```
R1(config)#router ospf 1
R1(config-router)#network 192.168.12.0 0.0.0.255 area 0
R1(config-router)#network 192.168.10.0 0.0.0.255 area 0
```

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 0

R2#show ip route ospf
O    192.168.10.0/24 [110/20] via 192.168.12.1, 00:03:21, FastEthernet0/0
```

OSPF Passive Interface

- 특정 인터페이스에 Hello 패킷 전송을 하지 않게 하기 위해 Passive-Interface 사용

```
R1#debug ip ospf hello
OSPF hello events debugging is on

OSPF: Send hello to 224.0.0.5 area 0 on FastEthernet0/1 from 192.168.10.254

OSPF: Send hello to 224.0.0.5 area 0 on FastEthernet0/0 from 192.168.12.1
```

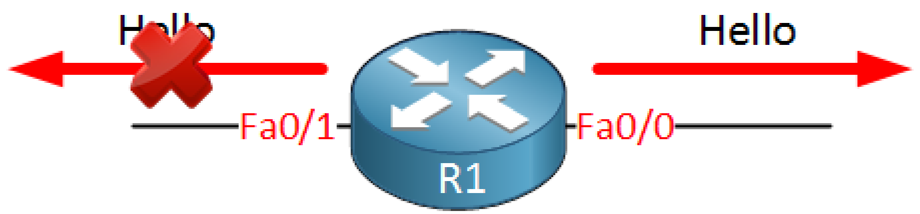


OSPF Passive Interface

- 특정 인터페이스에 Hello 패킷 전송을 하지 않게 하기 위해 Passive-Interface 사용

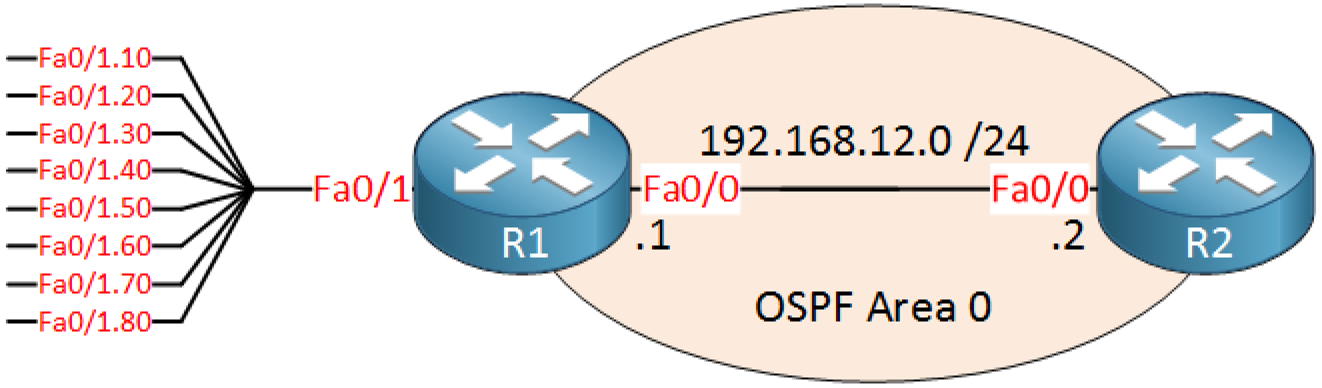
```
R1(config)#router ospf 1
R1(config-router)#passive-interface FastEthernet 0/1

R1#show ip protocols
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.12.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.10.0 0.0.0.255 area 0
    192.168.12.0 0.0.0.255 area 0
 Reference bandwidth unit is 100 mbps
  Passive Interface(s):
    FastEthernet0/1
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)
```

```
R1#
OSPF: Send hello to 224.0.0.5 area 0 on FastEthernet0/0 from
192.168.12.1
```



OSPF Passive Interface

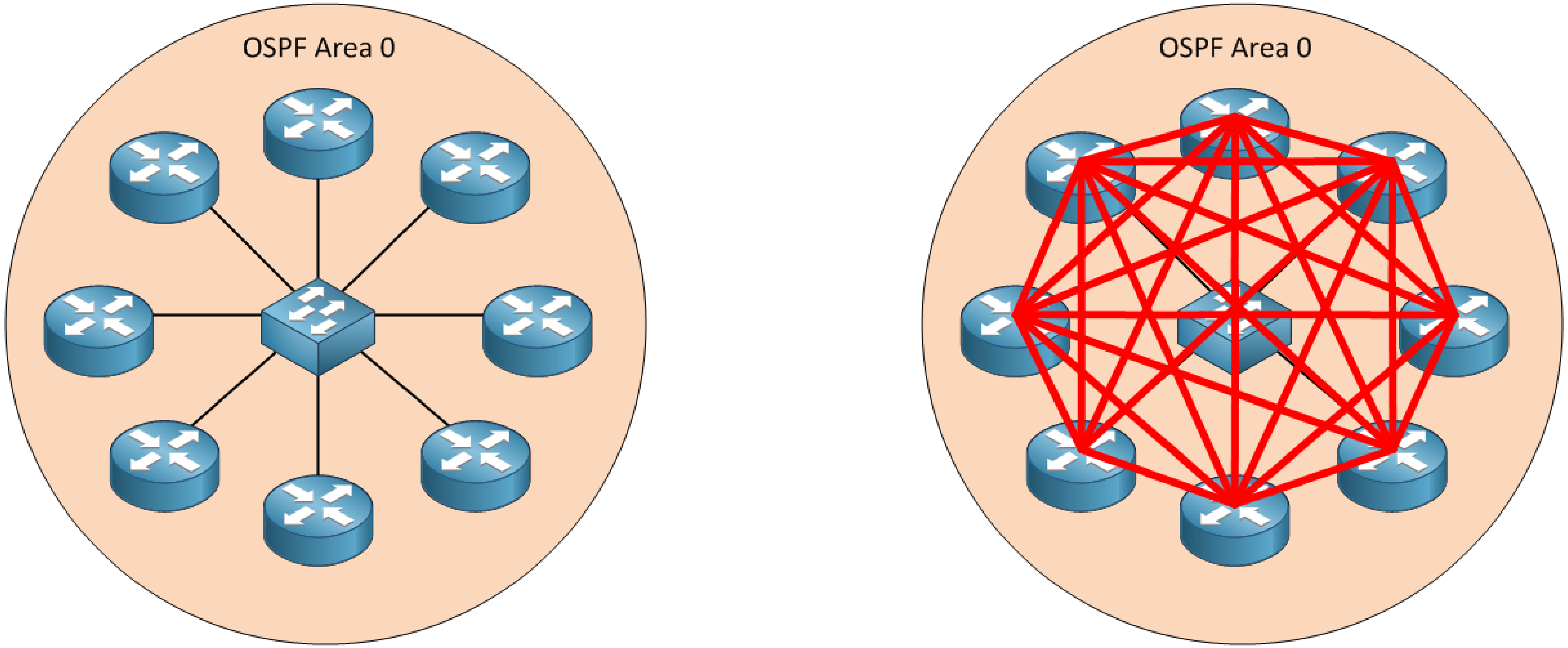
- 한 번에 많은 Passive-Interface 필요한 경우 White List 방식으로 적용 가능




```
R1(config)#router ospf 1
R1(config-router)#passive-interface default
R1(config-router)#no passive-interface FastEthernet 0/0
```

OSPF DR/BDR

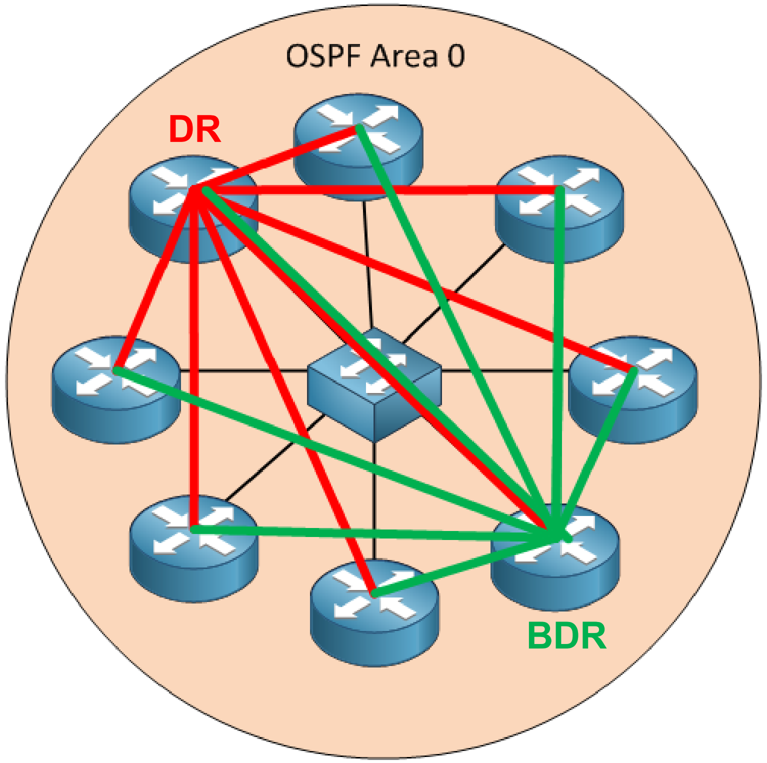
- Broadcast (Multi-Access) 환경에서 라우터가 많은 경우 각 라우터가 LS 정보를 다른 모든 라우터에 Floods 하는 것은 비효율적



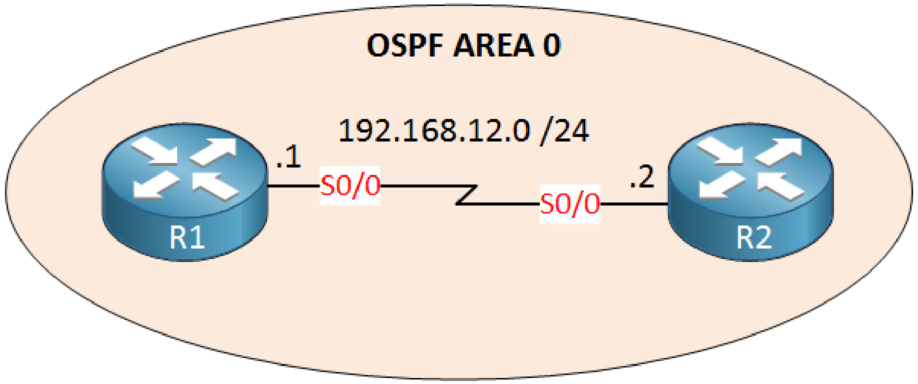
[8개의 라우터가 인접 관계를 맺고 LSDB를 구축 하는 경우]

OSPF DR/BDR

- 모든 라우터가 자신의 LS 정보를 하나의 DR 라우터에 보내고, DR 라우터가 다른 라우터에게 LS 정보 전달



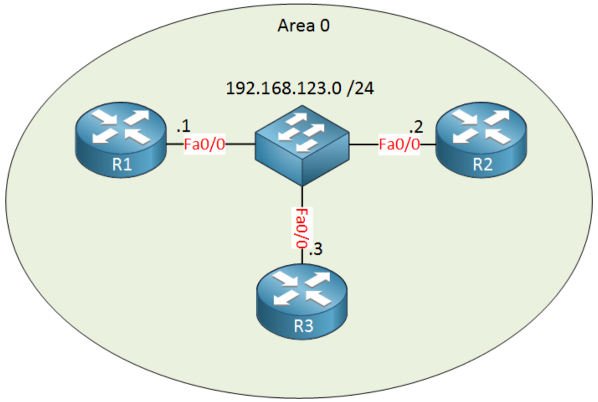
[DR/BDR 선출]

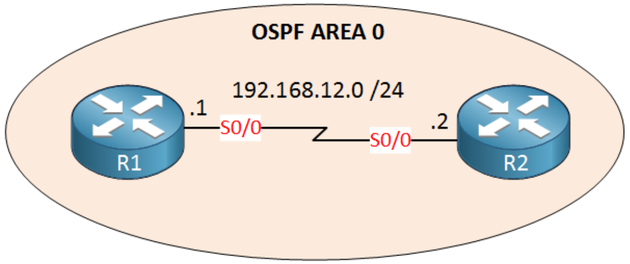


[DR/BDR 미선출]

OSPF DR/BDR Election

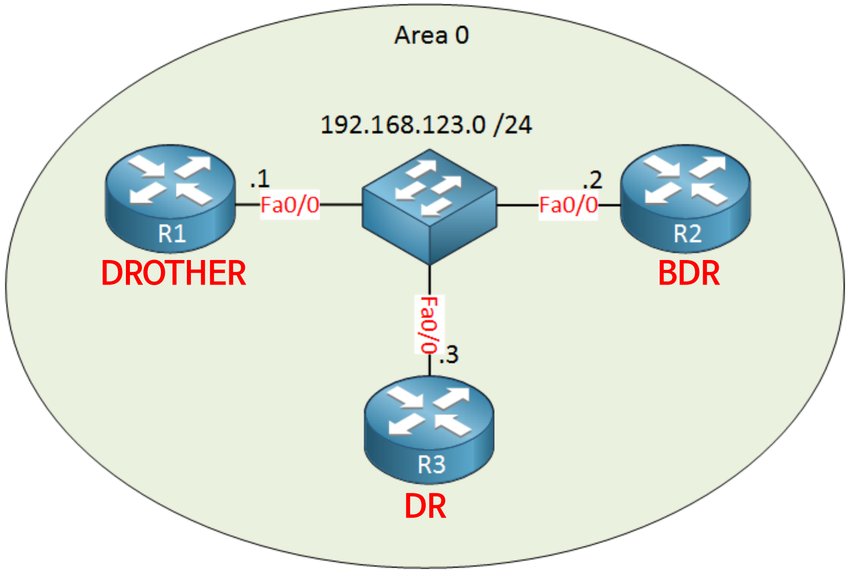
- OSPF는 인터페이스 유형을 확인 후 Multi-Access 환경에서 DR / BDR 선출
  - Multi-Access 환경 = 2개 이상의 Router를 갖고 있는 세그먼트
  - Ethernet 구간 : Multi-Access Network 환경 간주
  - Serial 구간 : Point-to-Point 환경 간주
- OSPF Neighbor 협상 과정에서 Hello 메시지 통해 OSPF Priority 교환
  - Default Priority : 1 (높은 값 우위)
  - Priority 0 장비는 절대 DR/BDR로 선출되지 않음





OSPF DR/BDR Election

- 스위치를 통해 FastEthernet 인터페이스로 연결된 Multi-Access 환경



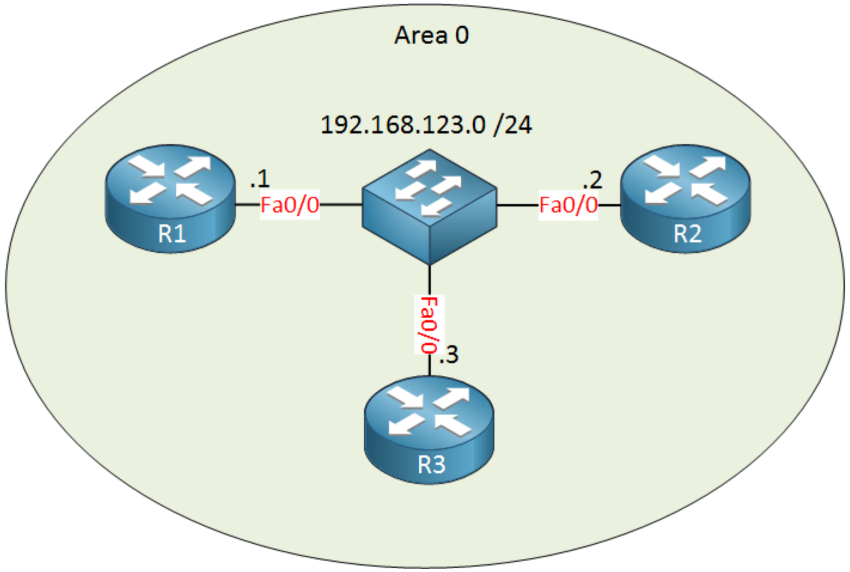

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.123.2     1   FULL/BDR        00:00:32    192.168.123.2 FastEthernet0/0
192.168.123.3     1   FULL/DR         00:00:31    192.168.123.3 FastEthernet0/0
```

```
R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.123.1     1   FULL/DROTHER    00:00:36    192.168.123.1 FastEthernet0/0
192.168.123.2     1   FULL/BDR        00:00:39    192.168.123.2 FastEthernet0/0
```

```
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.123.1     1   FULL/DROTHER    00:00:31    192.168.123.1 FastEthernet0/0
192.168.123.3     1   FULL/DR         00:00:32    192.168.123.3 FastEthernet0/0
```

OSPF DR/BDR Election

- OSPF Priority 값을 이용해 DR / BDR 변경
- Neighbor 재협상 필요




```
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip ospf priority 200
```

```
R3#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
```

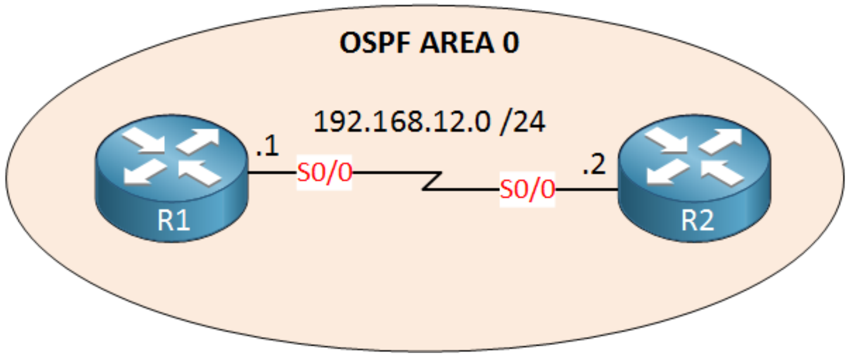
```
R2#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
```

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.123.2   1     FULL/DROTHER    00:00:36    192.168.123.2 FastEthernet0/0
192.168.123.3   1     FULL/BDR        00:00:30    192.168.123.3 FastEthernet0/0
```

```
R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.123.1   200   FULL/DR         00:00:30    192.168.123.1 FastEthernet0/0
192.168.123.2   1     FULL/DROTHER    00:00:31    192.168.123.2 FastEthernet0/0
```

OSPF DR/BDR Election

- Serial 인터페이스로 구성된 Point-to-Point 환경은 DR/BDR 선출 불필요



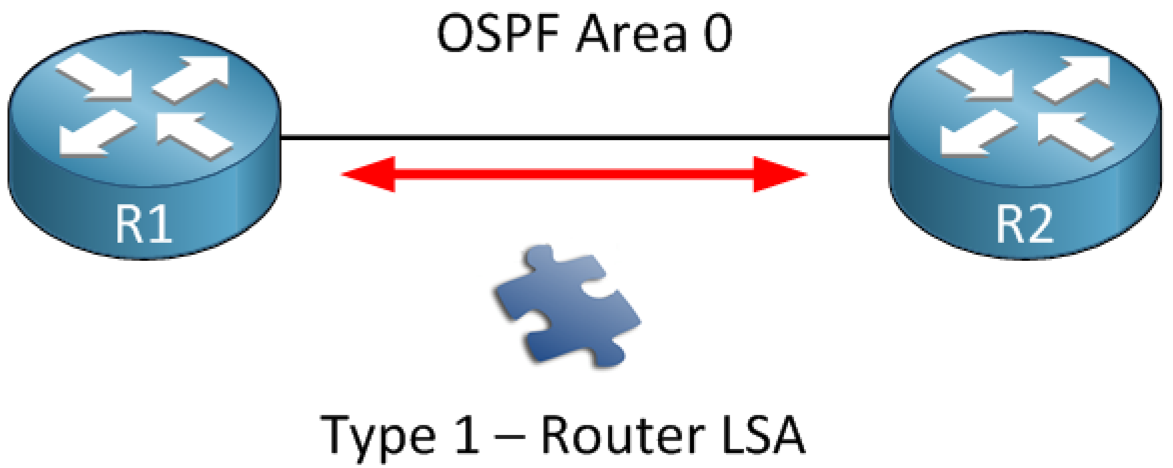

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.12.2      0   FULL/  -        00:00:36    192.168.12.2    Serial0/0
```

```
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.12.1      0   FULL/  -        00:00:34    192.168.12.1    Serial0/0
```

OSPF LSA Type

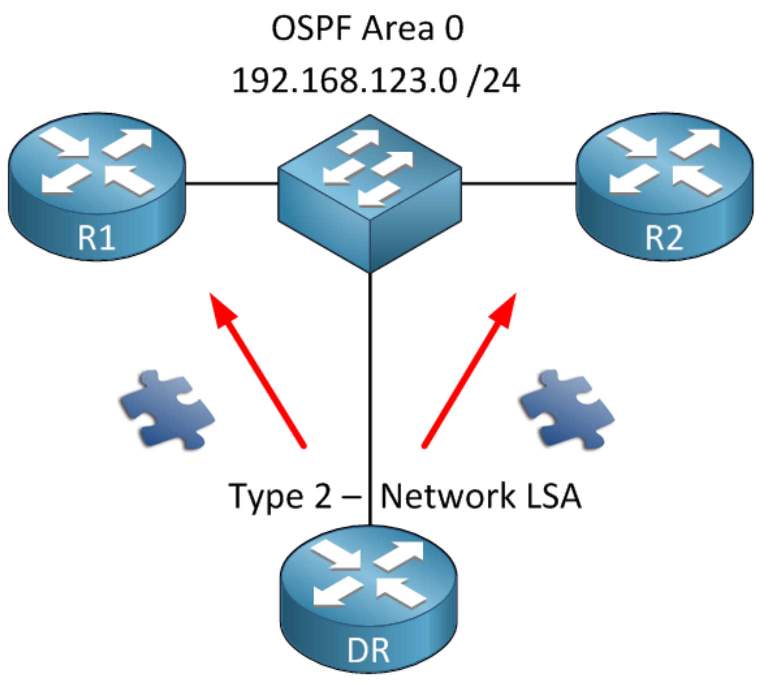
- OSPF는 LSA(Link State Advertisement)로 채워진 LSDB(Link State DB)를 사용

- Type 1 - Router LSA (Code : O)
  - 동일한 Area 내에서 Flood 되는 LSA
  - Connected Network를 포함한 LSA



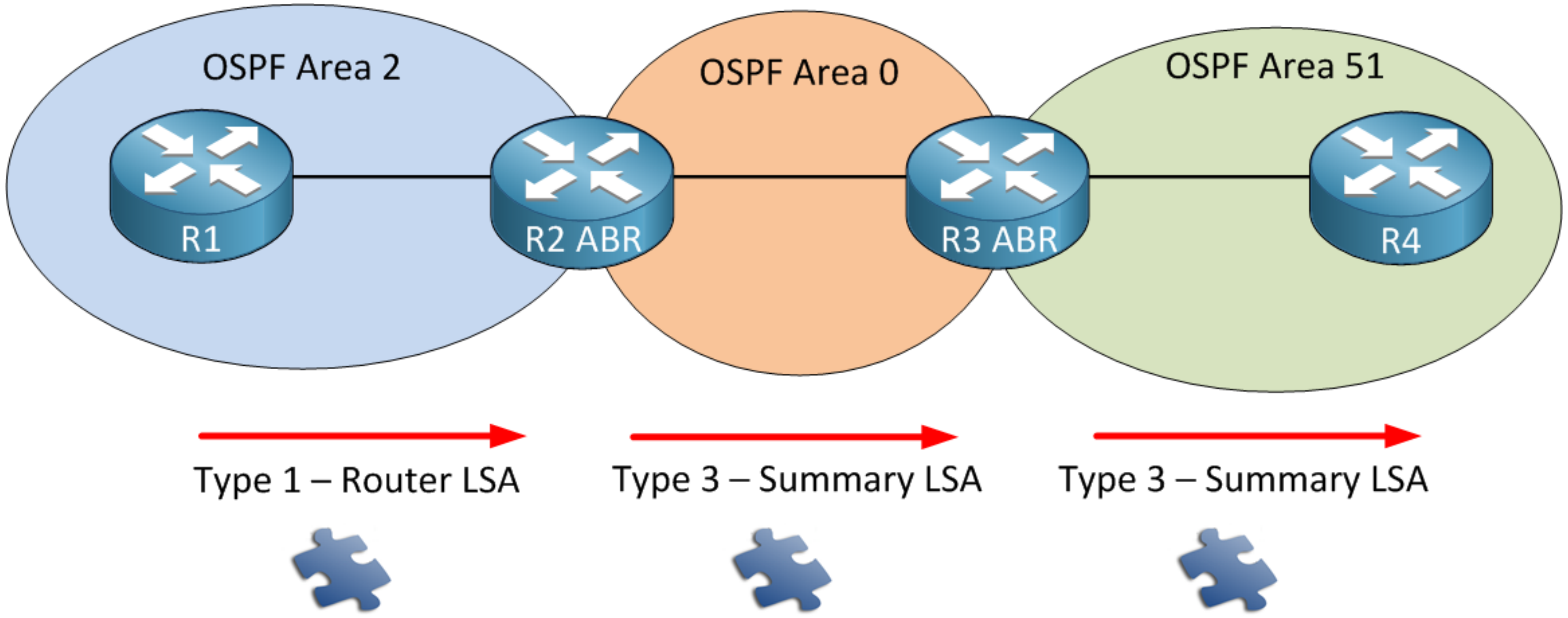
OSPF LSA Type

- Type 2 - Network LSA
  - Multi-Access 환경에서 선출된 DR에 의해 생성
  - DR의 Sub netmask와 DR에 연결된 라우터 목록을 포함
  - 동일한 Area 내에 Flood



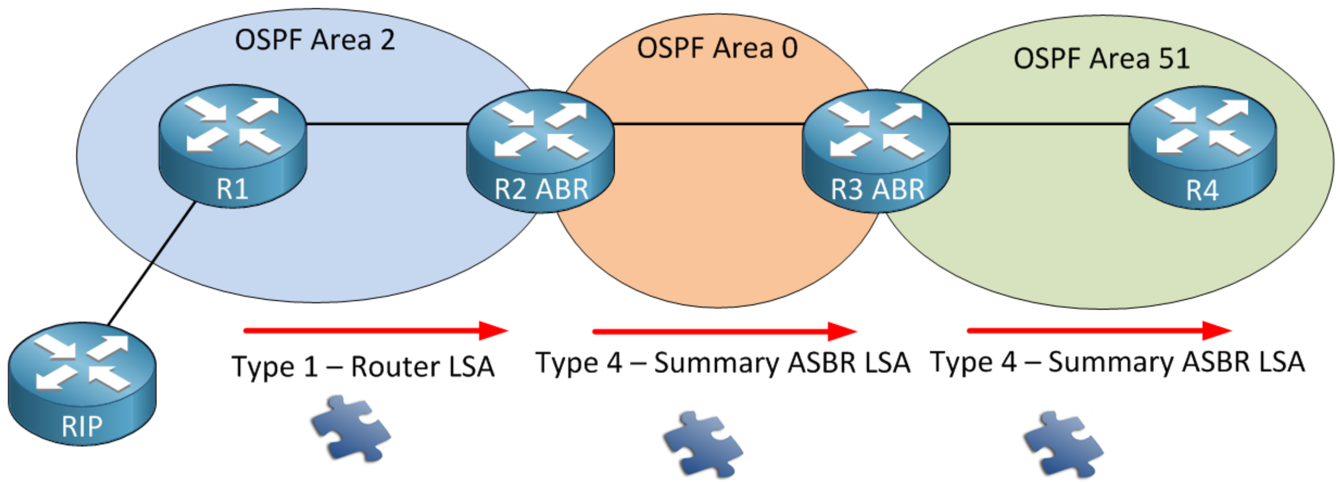
OSPF LSA Type

- Type 3 - Summary LSA (Code : O IA)
  - ABR에 의해 생성 되는 LSA
  - 다른 Area에 전달 하는 LSA
  - 이름은 Summary 지만 요약된 내용은 아님 (요약'**도**' 가능한 LSA)



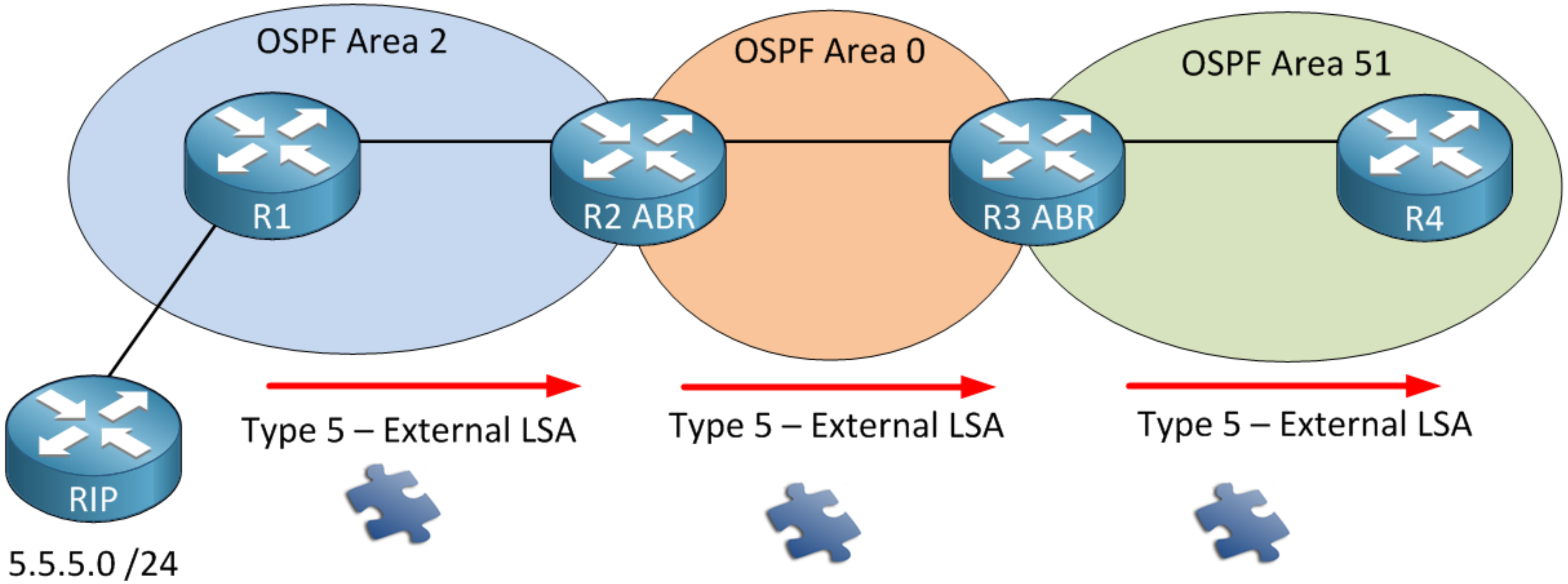
OSPF LSA Type

- Type 4 - Summary ASBR LSA
  - RIP 정보를 OSPF Area 내부로 전파해야 하는 R1은 ASBR 동작
  - R1은 동일한 Area내부에 Type1 or Type2 LSA 전파, 이를 수신한 R2는 Type-4로 변경해 Flood
  - OSPF 모든 라우터는 Type-4 LSA를 통해 ASBR 위치를 알게 됨



OSPF LSA Type

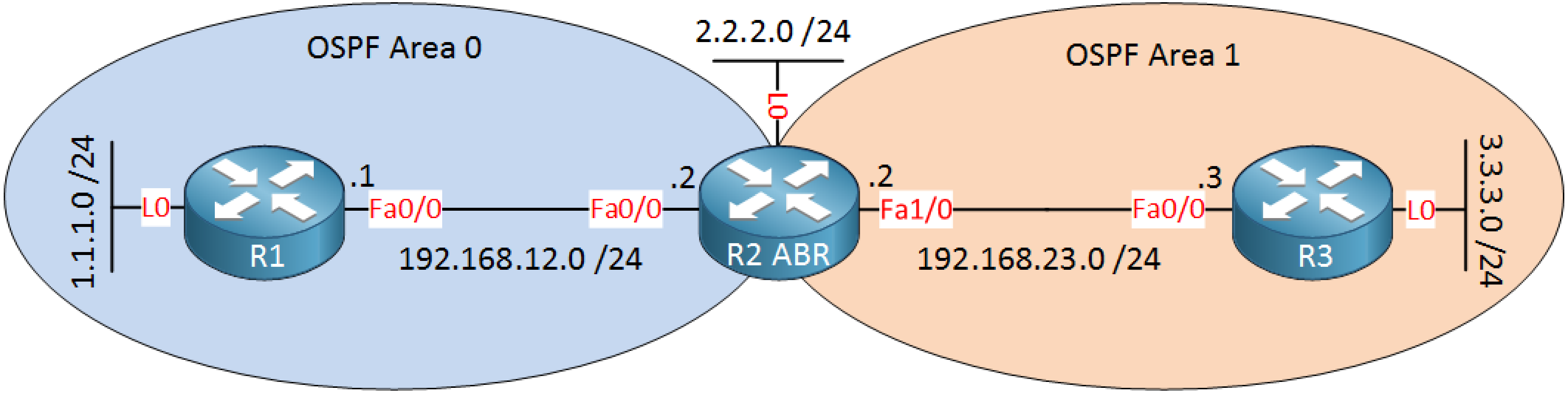
- Type 5 - external LSA (Code : O E1 , O E2)
  - ASBR(R1)에 의해 외부 경로가 OSPF 내부로 전파된 경로를 포함한 LSA
  - 다른 라우터는 Type-4를 통해 ASBR Router를 학습 하고 있어야 함



OSPF LSA Type Summary

- Type 1 - Router LSA
  - 라우터 LSA는 라우터가 위치한 각 지역에 대해 각 라우터에 의해 생성
- Type 2 - Network LSA
  - 네트워크 LSA는 DR에 의해 생성
- Type 3 - Summary LSA
  - 요약 LSA는 ABR에 의해 생성되고 다른 영역으로 전파
- Type 4 - Summary ASBR LSA
  - ASBR이 동작 하는 경우 다른 라우터는 ASBR을 위치 학습 필요
  - ASBR 라우터가 속하는 영역의 ABR이 생성해 다른 영역으로 전파
- Type 5 - External LSA
  - External LSA는 ASBR에 의해 생성 되고 , OSPF 내부에 전달 되기 원하는 외부 경로를 포함

OSPF LSA Type Verification



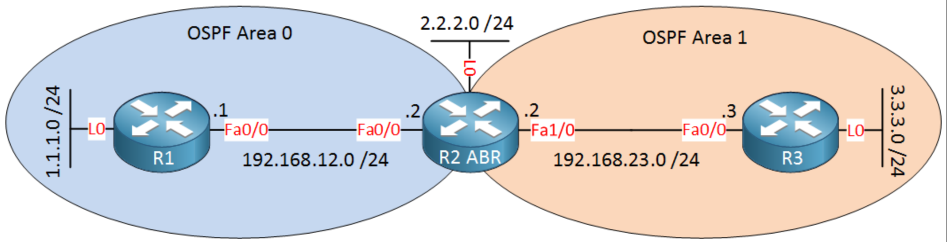

```
R1(config)#router ospf 1
R1(config-router)#network 192.168.12.0 0.0.0.255 area 0
R1(config-router)#network 1.1.1.0 0.0.0.255 area 0
```

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 0
R2(config-router)#network 192.168.23.0 0.0.0.255 area 1
```

```
R3(config)#router ospf 1
R3(config-router)#network 192.168.23.0 0.0.0.255 area 1
R3(config-router)#network 3.3.3.0 0.0.0.255 area 1
```

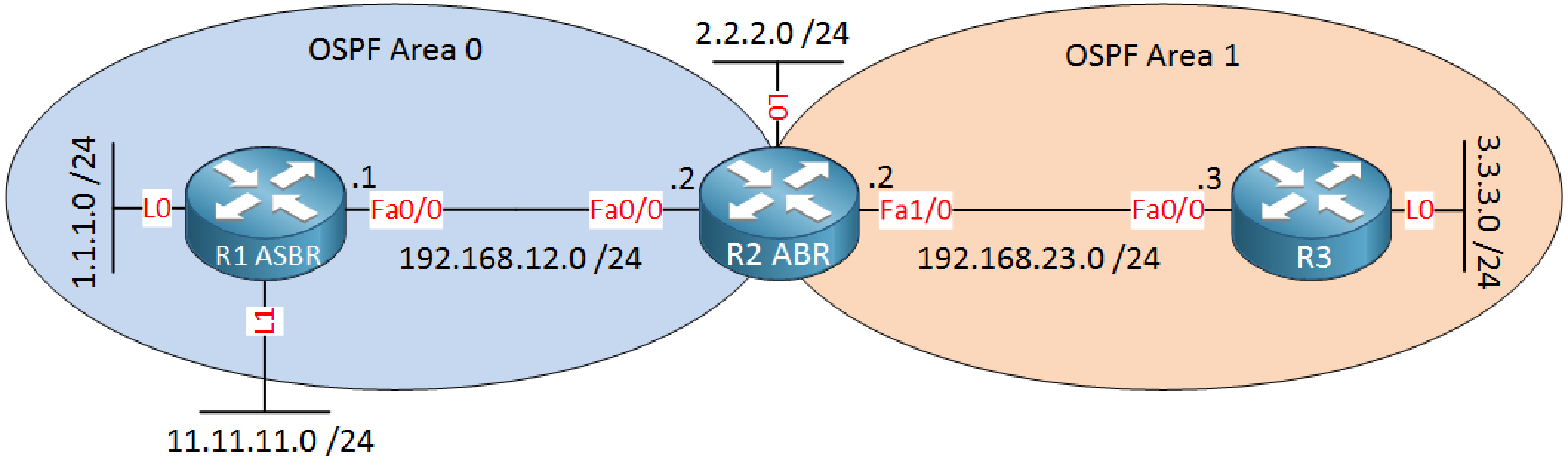
OSPF LSA Type Verification

- Type - 1,2,3




```
R1#show ip ospf database

            OSPF Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         30          0x80000003 0x004CD9 2
2.2.2.2         2.2.2.2         31          0x80000002 0x0048E9 1

                Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
192.168.12.2    2.2.2.2         31          0x80000001 0x008F1F

                Summary Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
3.3.3.3         2.2.2.2         17          0x80000001 0x00D650
192.168.23.0    2.2.2.2         66          0x80000001 0x00A70C
```

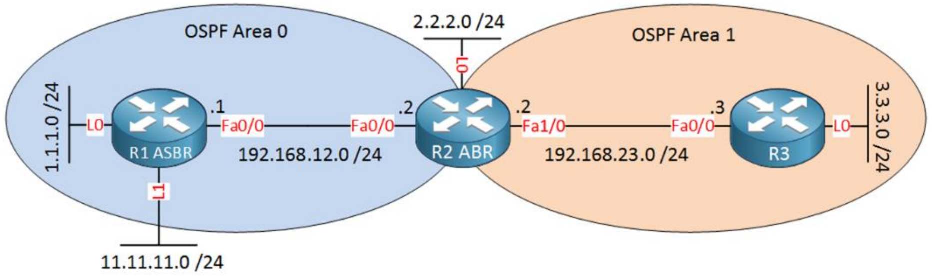
OSPF LSA Type Verification




```
R1(config)#interface loopback 1
R1(config-if)#ip address 11.11.11.11 255.255.255.0
R1(config-if)#exit

R1(config)#router ospf 1
R1(config-router)#redistribute connected subnets
```

OSPF LSA Type Verification

- Type - 4,5




```
R2#show ip ospf database | begin Type-5
                Type-5 AS External Link States

Link ID         ADV Router      Age         Seq#       Checksum Tag
11.11.11.0      1.1.1.1      36          0x80000001 0x000F44 0
```

```
R3#show ip ospf database | begin Summary
                Summary Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
1.1.1.1         2.2.2.2         149         0x80000001 0x0033FB
192.168.12.0    2.2.2.2         195         0x80000001 0x00219D

                Summary ASB Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
1.1.1.1         2.2.2.2         62          0x80000001 0x004DB9

                Type-5 AS External Link States

Link ID         ADV Router      Age         Seq#       Checksum Tag
11.11.11.0      1.1.1.1         68          0x80000001 0x000F44 0
```

OSPF Route Type

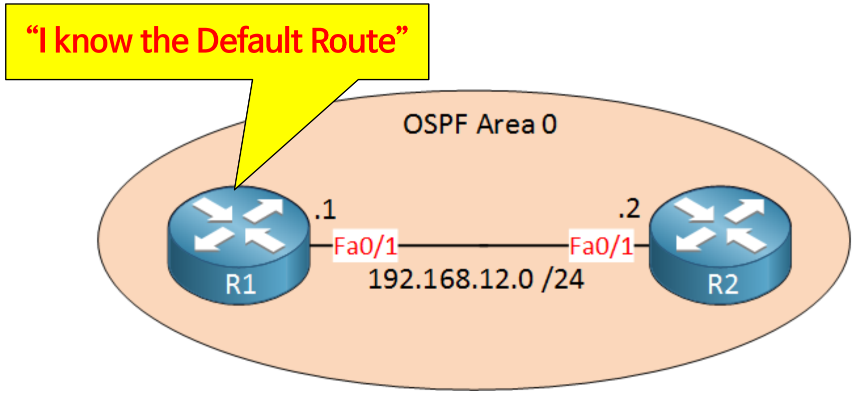
Route Type	Code	Priority	Description
Area 내부 경로	O	1	동일 Area 에 소속된 경로
Area 간 경로	O IA	2	다른 Area 에 소속된 경로
도메인 외부 경로	O E1	3	변동 Cost 값을 가지는 외부 경로
	O N1	4	변동 Cost 값을 가지는 NSSA 외부 경로
	O E2	5	고정 Cost 값을 가지는 외부 경로
	O N2	6	고정 Cost 값을 가지는 NSSA 외부 경로

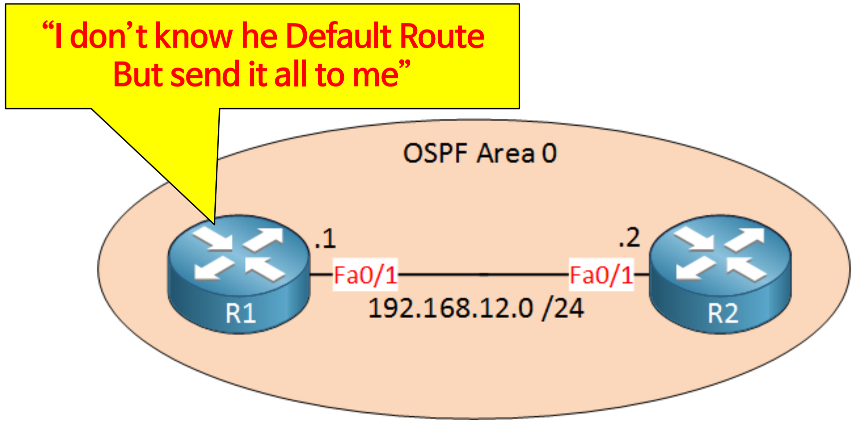
- OSPF 경로 AD는 모두 110 이지만 경로간 우선순위가 있어 최적 경로 경쟁 시 Cost 무시

OSPF Default Route

- OSPF를 이용해 Default Route 구성 방법
  - default-information originate
    - ✓ 라우팅 테이블에 Default Route가 있는 경우 광고
  - default-information originate always
    - ✓ 라우팅 테이블에 Default Route가 없어도 광고
- OSPF가 광고하는 Default Route의 Code는 O E2

OSPF Default Route






```
R1
router ospf 1
 network 192.168.12.0
 default-information originate
```

```
R1
router ospf 1
 network 192.168.12.0
 default-information originate always
```

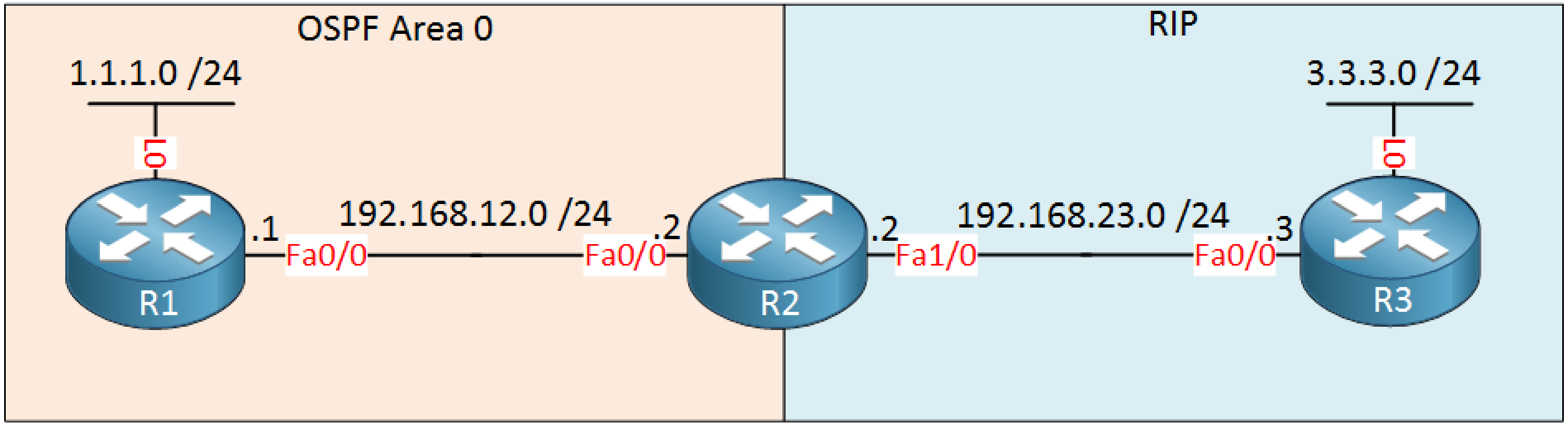
```
R2#show ip route ospf
O*E2 0.0.0.0/0 [110/1] via 192.168.12.1, 00:00:24, FastEthernet0/1
```

Introduction Redistribute

- 조직은 일반적으로 내부에 EIGRP,OSPF 등 단일 라우팅 프로토콜을 사용
  - 만약 OSPF를 운영하는 회사와 RIP을 사용하는 회사가 합병 된다면?
  - 만약 본사는 OSPF를 운영하지만 지사 장비는 구형장비라 OSPF를 지원하지 않고 오로지 RIP만 지원 한다면?

- 서로 다른 라우팅 프로토콜 간에 라우팅 정보를 교환 할 수 있는 방법이 필요 하고 이러한 방법을 **재배포/재분배** 라고 함
  - 그렇다면 Metric 값은? OSPF Cost 사용 , RIP Hop Count 사용 , EIGRP K-value 사용
  - 한쪽만 재분배를 하면 되는지?
  - 양쪽 모두 재분배를 하면 Loop이 발생하지 않는지? (재분배 받은 경로를 다시 재분배)

OSPF Redistribute



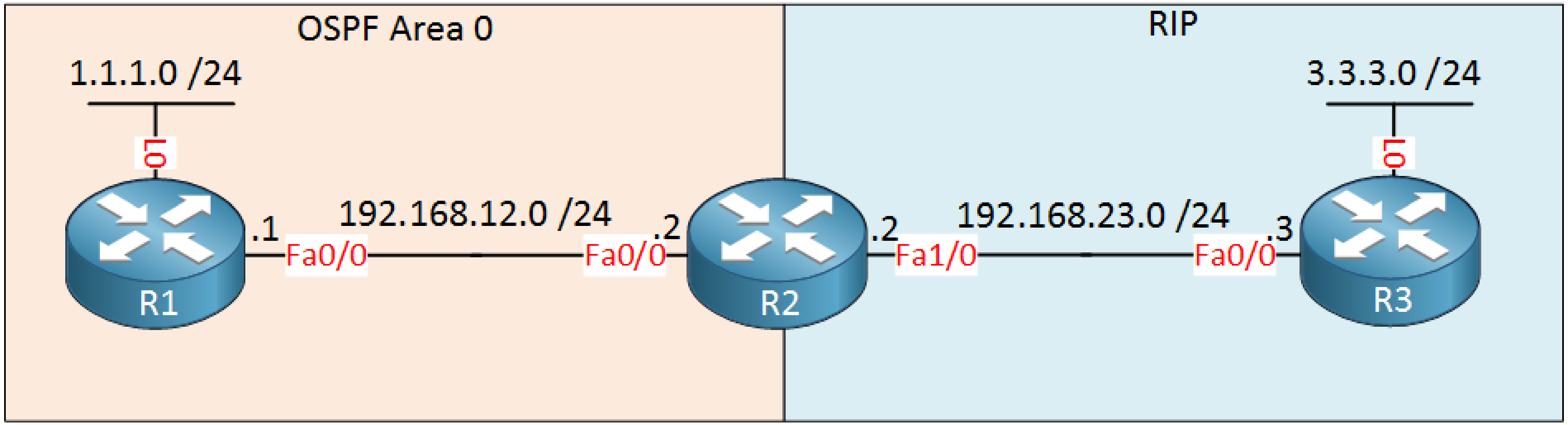

```
R1(config)#router ospf 1
R1(config-router)#network 1.1.1.0 0.0.0.255 area 0
R1(config-router)#network 192.168.12.0 0.0.0.255 area 0
```

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 0
```

```
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#no auto-summary
R2(config-router)#network 192.168.23.0
```

```
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#network 3.3.3.0
R3(config-router)#network 192.168.23.0
```

OSPF Redistribute

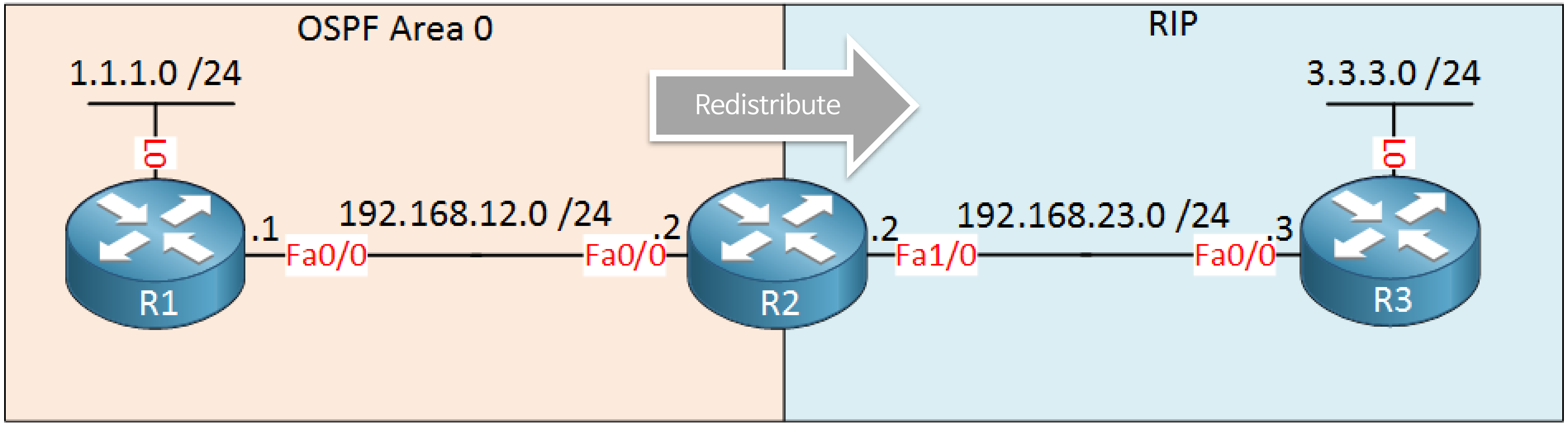



```
R1#show ip route

Gateway of last resort is not set

C    192.168.12.0/24 is directly connected, FastEthernet0/0
     1.0.0.0/24 is subnetted, 1 subnets
C       1.1.1.0 is directly connected, Loopback0
```

```
R3#show ip route

Gateway of last resort is not set

     3.0.0.0/24 is subnetted, 1 subnets
C       3.3.3.0 is directly connected, Loopback0
C    192.168.23.0/24 is directly connected, FastEthernet0/0
```

```
R2#show ip route

Gateway of last resort is not set

C    192.168.12.0/24 is directly connected, FastEthernet0/0
     1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/2] via 192.168.12.1, 00:11:05, FastEthernet0/0
     3.0.0.0/24 is subnetted, 1 subnets
R       3.3.3.0 [120/1] via 192.168.23.3, 00:00:20, FastEthernet1/0
C    192.168.23.0/24 is directly connected, FastEthernet1/0
```

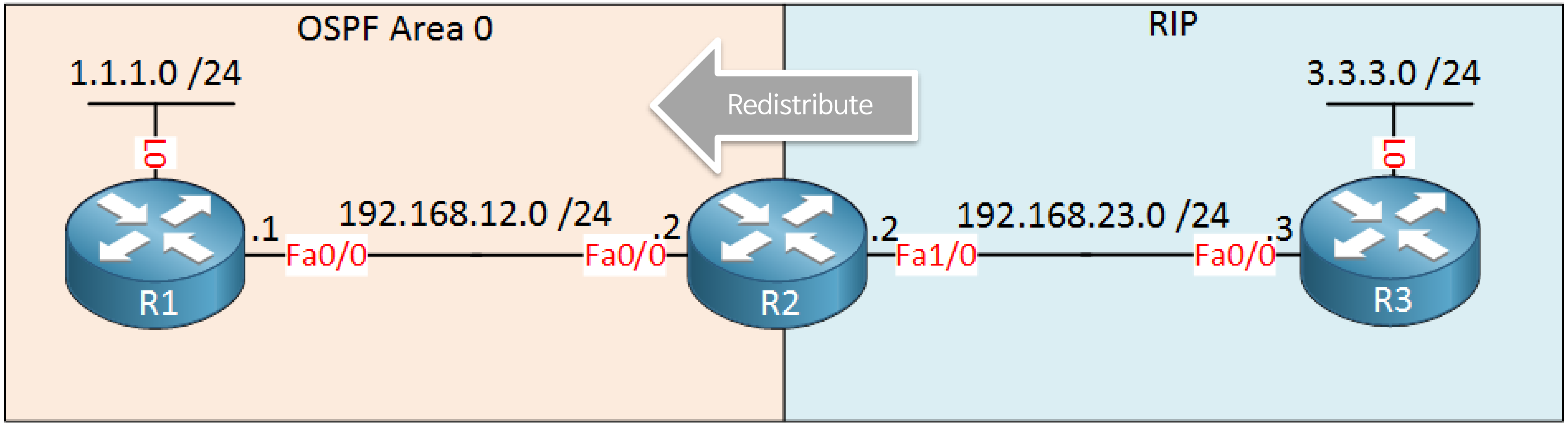
OSPF Redistribute




```
R2(config)#router rip
R2(config-router)#redistribute ospf 1 metric 5
```

```
R3#show ip route rip
R    192.168.12.0/24 [120/5] via 192.168.23.2, 00:00:00, FastEthernet0/0
     1.0.0.0/32 is subnetted, 1 subnets
R       1.1.1.1 [120/5] via 192.168.23.2, 00:00:00, FastEthernet0/0
```

OSPF Redistribute

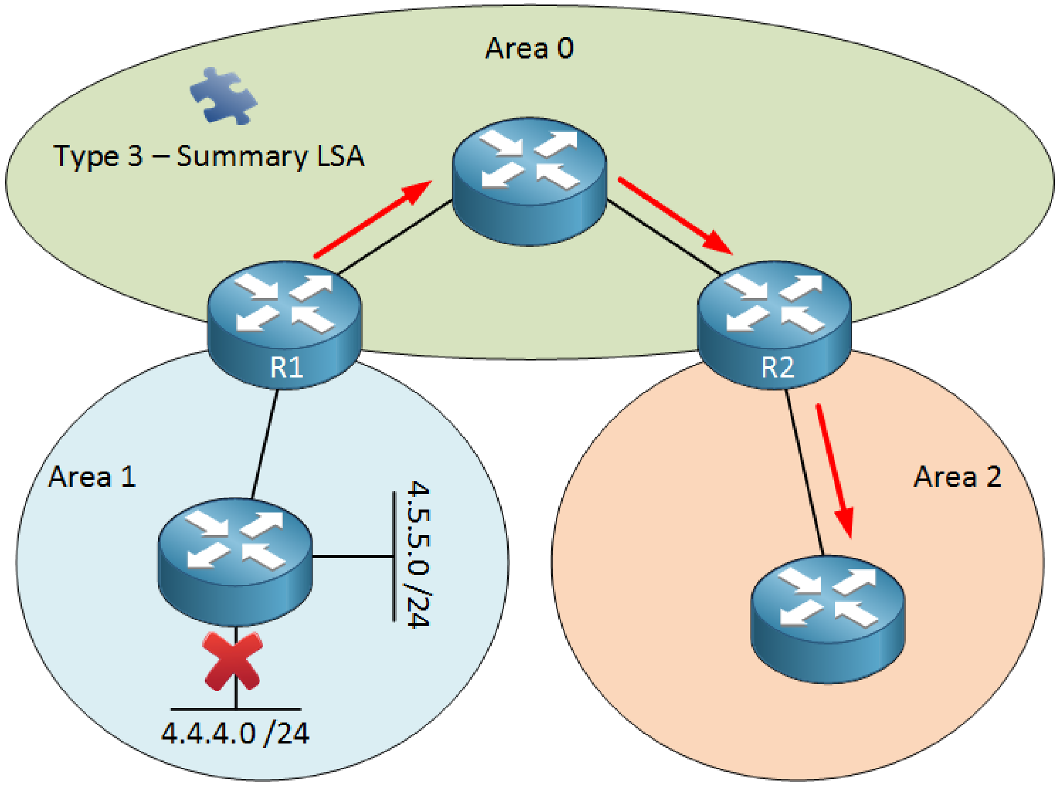



```
R2(config)#router ospf 1
R2(config-router)#redistribute rip subnets
```

```
R1#show ip route ospf
     3.0.0.0/24 is subnetted, 1 subnets
O E2    3.3.3.0 [110/20] via 192.168.12.2, 00:00:21, FastEthernet0/0
O E2 192.168.23.0/24 [110/20] via 192.168.12.2, 00:00:21, FastEthernet0/0
```

OSPF Summarization

- OSPF는 Area 내 경로 요약 불가능
- ABR 또는 ASBR 통해 경로 요약 가능 (Type-3 , Type-5)



1) Area1 내부 링크 변화에 따른 LAS가 모든 Area에 Flooding 됨.
2) 모든 라우터는 LSDB update 후 SPF 재실행 필요.

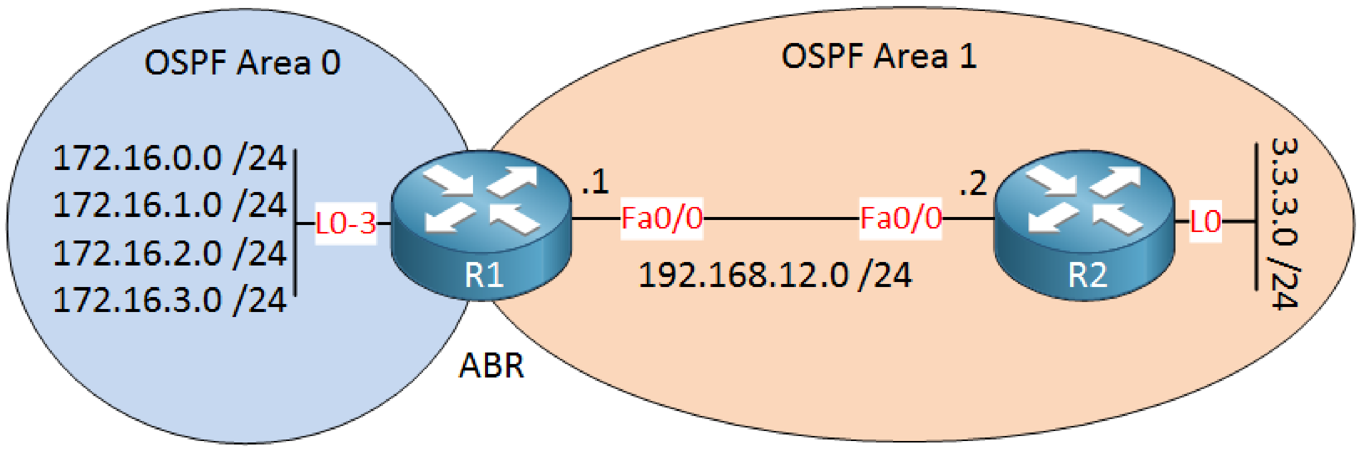
R1 (ABR)이 4.0.0.0/8 같은 경로를 전파 하면?

- 링크 변화에 따른 LSA Flooding 발생 감소
- 즉. LSDB Update 및 SPF 재실행 불필요

OSPF Summarization

- Type-3 / 5 Summarization
  - 요약 경로는 요약 범위에 속하는 서브넷이 하나 이상 있는 경우에만 광고
  - 요약 경로는 요약 범위 내에 속하는 서브넷 중 비용이 가장 낮은 서브넷의 비용을 사용 (only Type-3)
  - 요약 경로를 생성하는 ABR / ASBR은 요약 경로 전파를 위한 null0 경로 생성
  - OSPF는 클래스 없는 라우팅 프로토콜이므로 원하는 서브넷 마스크 선택 가능

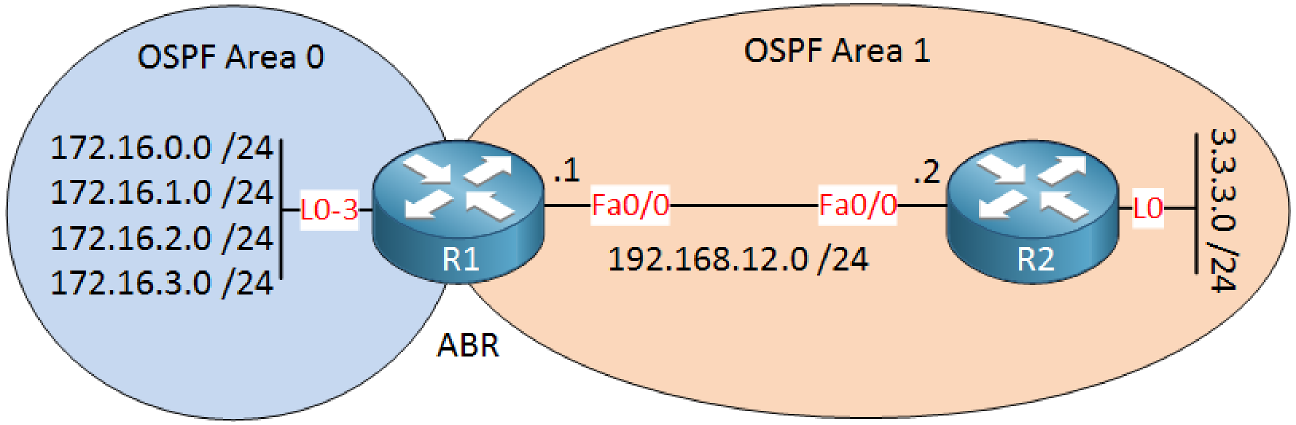
OSPF Summarization




```
R1(config)#router ospf 1
R1(config-router)#network 172.16.0.0 0.0.3.255 area 0
R1(config-router)#network 192.168.12.0 0.0.0.255 area 1
```

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 1
```

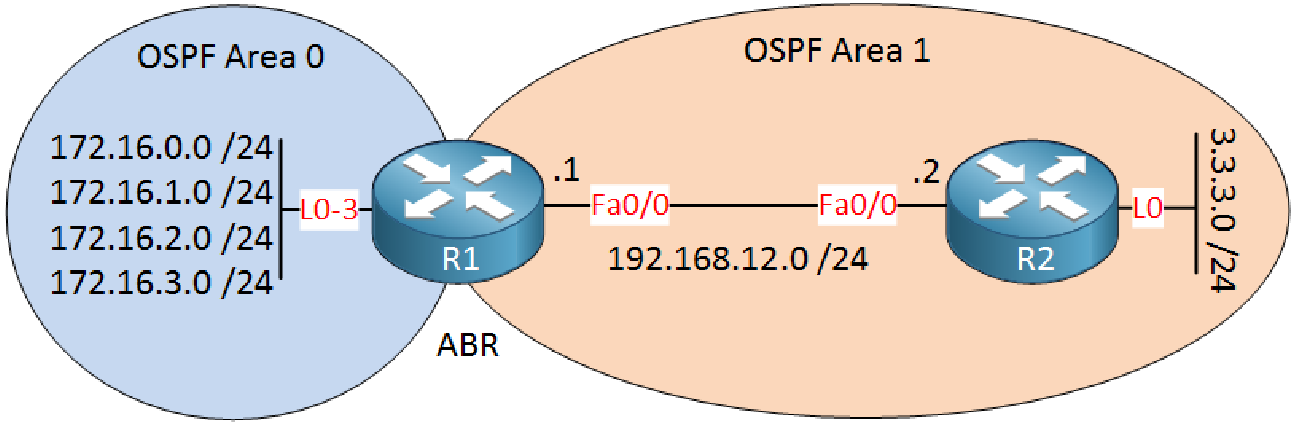
OSPF Summarization




```
R2#show ip route ospf
     172.16.0.0/32 is subnetted, 4 subnets
O IA    172.16.1.1 [110/2] via 192.168.12.1, 00:08:04, FastEthernet0/0
O IA    172.16.0.1 [110/2] via 192.168.12.1, 00:08:04, FastEthernet0/0
O IA    172.16.3.1 [110/2] via 192.168.12.1, 00:08:04, FastEthernet0/0
O IA    172.16.2.1 [110/2] via 192.168.12.1, 00:08:04, FastEthernet0/0
```

```
R2#show ip ospf database | begin Summary
                Summary Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
172.16.0.1      172.16.3.1      542         0x80000003 0x005269
172.16.1.1      172.16.3.1      542         0x80000003 0x004773
172.16.2.1      172.16.3.1      542         0x80000003 0x003C7D
172.16.3.1      172.16.3.1      542         0x80000003 0x003187
```

OSPF Summarization

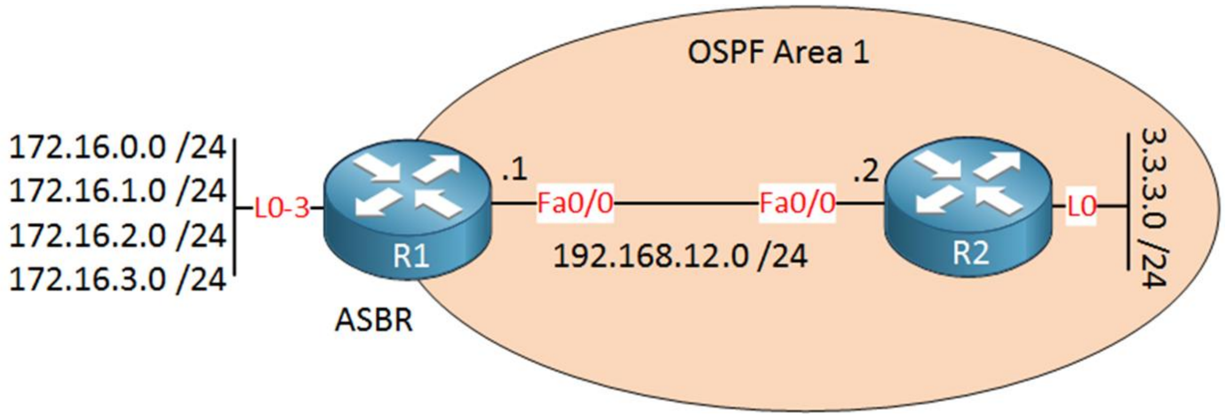



```
R1(config)#router ospf 1
R1(config-router)#area 0 range 172.16.0.0 255.255.0.0
```

```
R2#show ip route ospf
O IA 172.16.0.0/16 [110/2] via 192.168.12.1, 00:03:26, FastEthernet0/0
```

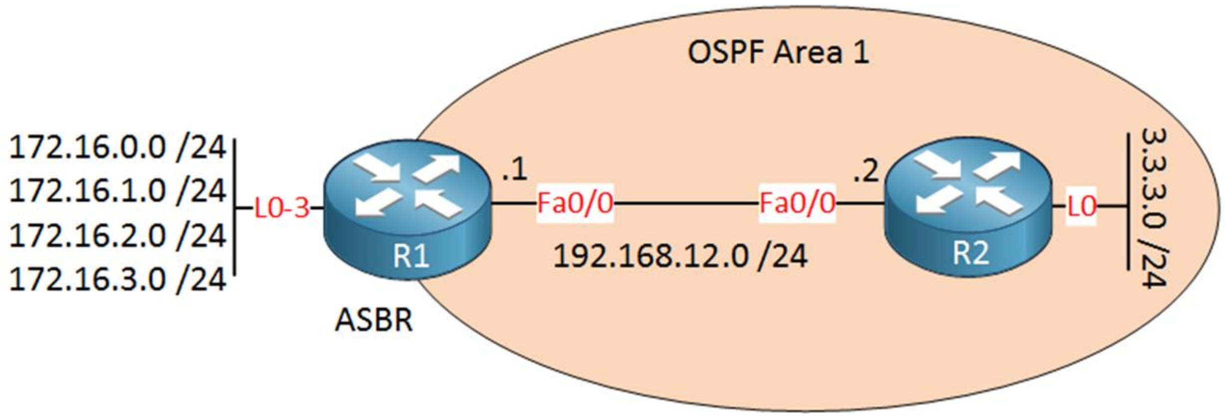
```
R2#show ip ospf database | begin Summary
                     Summary Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
172.16.0.0      172.16.3.1      219         0x80000001 0x00605E
```

OSPF Summarization




```
R1(config)#router ospf 1
R1(config-router)#no network 172.16.0.0 0.0.3.255 area 0
R1(config-router)#redistribute connected subnets
```

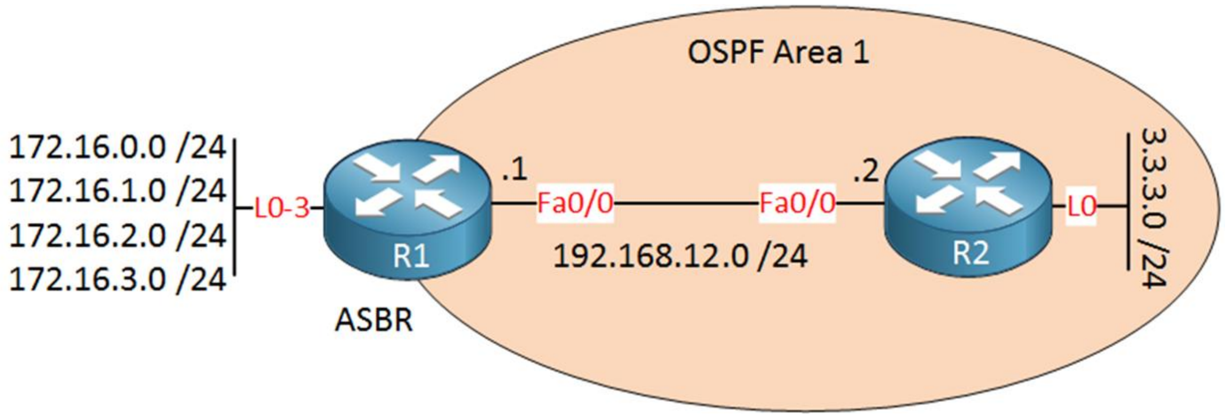
OSPF Summarization




```
R2#show ip route ospf
     172.16.0.0/24 is subnetted, 4 subnets
O E2    172.16.0.0 [110/20] via 192.168.12.1, 00:01:07, FastEthernet0/0
O E2    172.16.1.0 [110/20] via 192.168.12.1, 00:01:07, FastEthernet0/0
O E2    172.16.2.0 [110/20] via 192.168.12.1, 00:01:07, FastEthernet0/0
O E2    172.16.3.0 [110/20] via 192.168.12.1, 00:01:07, FastEthernet0/0
```

```
R2#show ip ospf database | begin Type-5
                Type-5 AS External Link States

Link ID         ADV Router      Age         Seq#       Checksum Tag
172.16.0.0      172.16.3.1      91          0x80000001 0x00B46E 0
172.16.1.0      172.16.3.1      91          0x80000001 0x00A978 0
172.16.2.0      172.16.3.1      91          0x80000001 0x009E82 0
172.16.3.0      172.16.3.1      91          0x80000001 0x00938C 0
```

OSPF Summarization



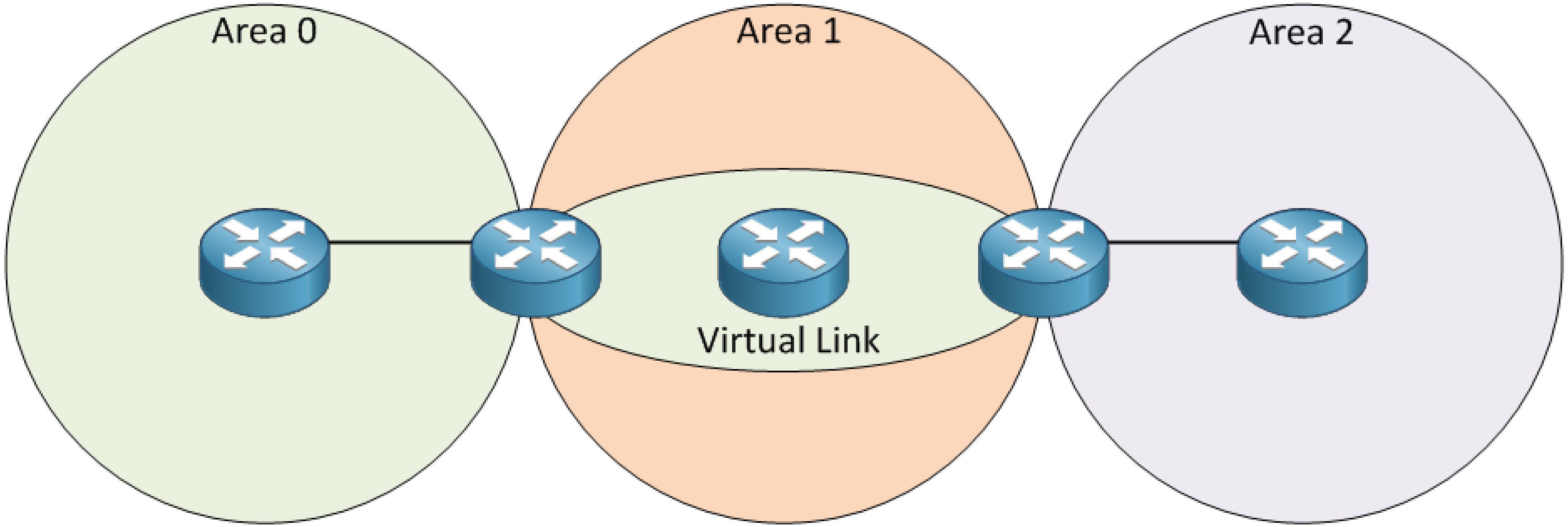

```
R1(config)#router ospf 1
R1(config-router)#summary-address 172.16.0.0 255.255.0.0
```

```
R2#show ip route ospf
O E2 172.16.0.0/16 [110/20] via 192.168.12.1, 00:00:17, FastEthernet0/0
```

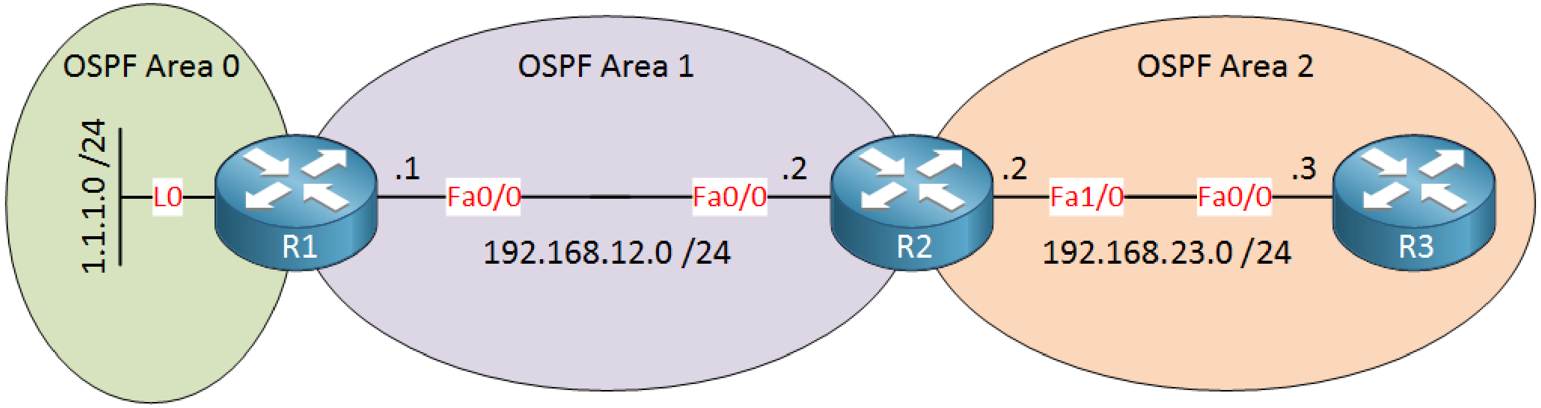
```
R2#show ip ospf database | begin Type-5
                Type-5 AS External Link States

Link ID         ADV Router      Age         Seq#       Checksum Tag
172.16.0.0      172.16.3.1      38          0x80000002 0x00B26F 0
```

OSPF Virtual Link

- OSPF 의 모든 Area는 Backbone Area와 연결 필수
- 만약 Backbone Area와 연결할 수 없는 상황이라면? ➔ Virtual Link 필요



OSPF Virtual Link

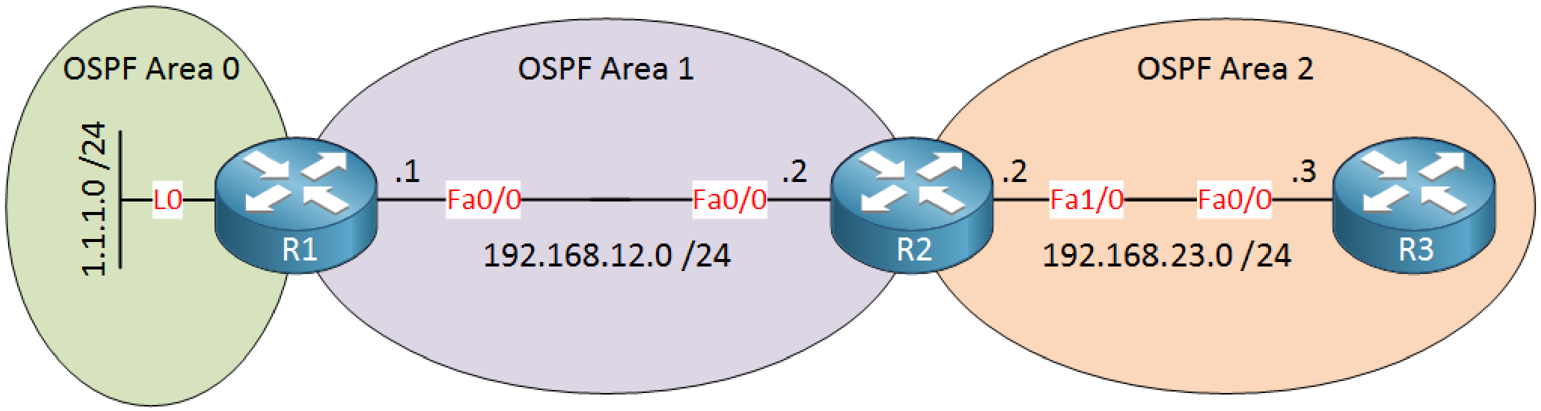



```
R1(config)#router ospf 1
R1(config-router)#network 1.1.1.0 0.0.0.255 area 0
R1(config-router)#network 192.168.12.0 0.0.0.255 area 1
```

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 0.0.0.255 area 1
R2(config-router)#network 192.168.23.0 0.0.0.255 area 2
```

```
R3(config)#router ospf 1
R3(config-router)#network 192.168.23.0 0.0.0.255 area 2
```

OSPF Virtual Link



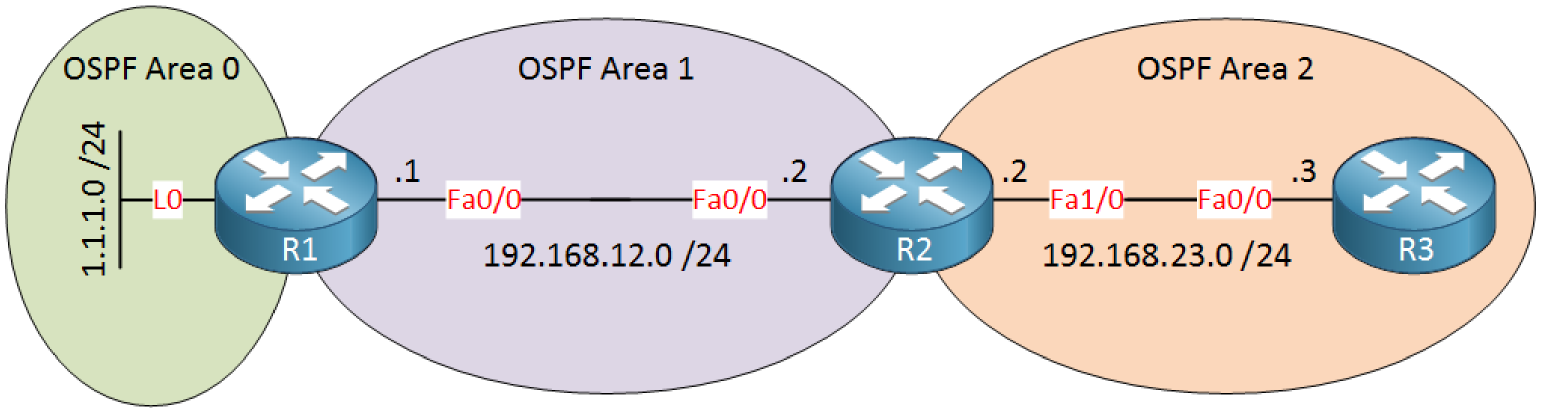

```
R1(config)#router ospf 1
R1(config-router)#area 1 virtual-link 192.168.23.2
```

```
R2(config)#router ospf 1
R2(config-router)#area 1 virtual-link 1.1.1.1
```

```
R1# %OSPF-5-ADJCHG  Process 1, Nbr 192.168.23.2 on OSPF_VL0 from LOADING to FULL, Loading Done
```

```
R2# %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on OSPF_VL0 from LOADING to FULL, Loading Done
```

OSPF Virtual Link




```
R1#show ip ospf virtual-links
Virtual Link OSPF_VL0 to router 192.168.23.2 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 1, via interface FastEthernet0/0, Cost of using 1
```

```
R2#show ip ospf virtual-links
Virtual Link OSPF_VL0 to router 1.1.1.1 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 1, via interface FastEthernet0/0, Cost of using 1
```

OSPF Virtual Link

```
R1#show ip ospf database

            OSPF Router with ID (1.1.1.1) (Process ID 1)

                      Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         189         0x80000004 0x00E333 2
192.168.23.2    192.168.23.2    1     (DNA) 0x80000002 0x009816 1
```

```
R2#show ip ospf database

            OSPF Router with ID (192.168.23.2) (Process ID 1)

                      Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         1     (DNA) 0x80000004 0x00E333 2
192.168.23.2    192.168.23.2    159         0x80000002 0x009816 1
```